

Casual Document for Probabilistic Principal Component Analysis MLE Asymptotic Normality

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Chapter 1

Low Dimensional Asymptotic Normality

1.1 Basic Setup

Given a "high dimensional" data set $\{\mathbf{x}_i\}_{i=1}^n$ where $\mathbf{x}_i \in \mathbb{R}^p$ for all $i = 1, \dots, n$, the statistician fixes a positive integer $q < p$ and wishes to find the q principal components. Assuming that the data is centred (that is, with mean 0), the probabilistic principal component analysis model is a statistical model that treats $\mathbf{x}_i$'s as realizations of independent random variables $\mathbf{X}_i = \mathbf{W}\mathbf{Z}_i + \boldsymbol{\epsilon}_i$, where $\mathbf{W}$ is an unknown $p \times q$ loading matrix of full rank, $\mathbf{Z}_i \sim N_q(\mathbf{0}, \mathbf{I}_q)$ is the latent normal variable, and $\boldsymbol{\epsilon}_i \sim N_p(\mathbf{0}, \sigma^2 \mathbf{I}_p)$ is the Gaussian noise with unknown standard deviation $\sigma > 0$, and $(\mathbf{Z}_1, \dots, \mathbf{Z}_n, \boldsymbol{\epsilon}_1, \dots, \boldsymbol{\epsilon}_n)$ are all independent (for now). Put $\theta = (\mathbf{W}_\theta, \sigma_\theta^2)$ for the unknown parameter to estimate and Θ for all possible choices of θ . Θ is thus the collection of all tuples $(\mathbf{W}, \sigma^2)$ such that $\mathbf{W}$ is a real $p \times q$ matrix with $\text{rank}(\mathbf{W}) = q$ and $\sigma^2 \in (0, \infty)$. Given that the collection of all real $p \times q$ matrix with rank q is an open subset of the Euclidean space $\mathbb{R}^{pq}$, we consider Θ as an open subset of the Euclidean space $\mathbb{R}^{pq+1}$ and our statistical model is

$$\mathcal{P} = \{N_p(\mathbf{0}, \mathbf{W}_\theta \mathbf{W}_\theta^T + \sigma_\theta^2 \mathbf{I}_p)\}_{\theta \in \Theta}. \quad (1.1)$$

To resolve the non-identifiability of $\mathcal{P}$, we introduce the quotient manifold structure. For $(\mathbf{A}, s) \in \Theta$ and $(\mathbf{B}, t) \in \Theta$, we define $(\mathbf{A}, s) \sim (\mathbf{B}, t)$ to mean that they give rise to the same covariance matrix, namely when $\mathbf{A}\mathbf{A}^T + s\mathbf{I}_p = \mathbf{B}\mathbf{B}^T + t\mathbf{I}_p$. The equivalence classes are denoted by $[(\mathbf{A}, s)]$ or more briefly $[\mathbf{A}, s]$. Unless otherwise mentioned, $(\mathbf{A}, s) \in \Theta$ is the default representative we pick when we write an equivalence class as $[\mathbf{A}, s]$. Writing $O(q)$ for the collection of all $q \times q$ real orthogonal matrices, it was shown in Datta's thesis that $[\mathbf{A}, s] = \{(\mathbf{A}\mathbf{Q}, s) \mid \mathbf{Q} \in O(q)\}$. Therefore, $M := \Theta / \sim$ can be identified as the orbits of the right action of $O(q)$ on Θ and, given that this Lie group action is smooth, proper, and free, the quotient manifold theorem (see [Lee(2013)], Theorem 21.10) shows that there is a unique smooth structure making M a manifold of dimension $pq + 1 - \frac{q(q-1)}{2}$ and the map $\pi : \Theta \rightarrow M$ defined by

$$\pi((\mathbf{A}, s)) = [\mathbf{A}, s] \quad (1.2)$$

is a smooth submersion. Moreover, this action is verticle, transitive on fibers and is isomertic with respect to the Euclidean metric. Thus, there exists a unique metric g on M making (M, g) a Riemannian manifold and π a Riemannian submersion (see [Lee(2018)], Theorem 2.28). In more concrete terms, if g^Θ is the standard Euclidean metric given by

$$g_{(\mathbf{A}, s)}^\Theta((\mathbf{V}, c), (\mathbf{U}, b)) = \text{tr}(\mathbf{V}\mathbf{U}^T) + cb, \quad (1.3)$$

whenever $(\mathbf{A}, s) \in \Theta$, $(\mathbf{V}, c), (\mathbf{U}, b) \in T_{(\mathbf{A}, s)}\Theta \cong \mathbb{R}^{p \times q} \times \mathbb{R}$, then

$$g_{[\mathbf{A}, s]}(\mathbf{v}, \mathbf{h}) = g_{[\mathbf{A}, s]}(d\pi_{(\mathbf{A}, s)}((\mathbf{V}, c)), d\pi_{(\mathbf{A}, s)}((\mathbf{U}, b))) = g_{(\mathbf{A}, s)}^\Theta((\mathbf{V}, c), (\mathbf{U}, b)), \quad (1.4)$$

whenever $\mathbf{v}, \mathbf{h} \in T_{[\mathbf{A}, s]}M$ lifting to $(\mathbf{V}, c)$ and $(\mathbf{U}, b)$ respectively. For later convenience, let us define two functions $\Phi : \Theta \rightarrow \mathbb{R}^{p \times p}$ and $\Sigma : M \rightarrow \mathbb{R}^{p \times p}$ by setting, for every $[\mathbf{A}, s] \in M$:

$$\Sigma([\mathbf{A}, s]) := \Phi((\mathbf{A}, s)) := \mathbf{A}\mathbf{A}^T + s\mathbf{I}_p. \quad (1.5)$$

In other words, $\Sigma \circ \pi = \Phi$ gives the covariance matrix. (1.1) is then reparametrized as

$$\mathcal{P} = \{N_p(\mathbf{0}, \Sigma([\mathbf{A}, s]))\}_{[\mathbf{A}, s] \in M} \quad (1.6)$$

and this is the identifiable model we shall work with.

Finally, throughout the entire document, $\|\mathbf{A}\|$ will denote the spectral norm of a matrix $\mathbf{A}$, which is defined as the largest singular value: the largest eigenvalue of $(\mathbf{A}^T \mathbf{A})^{\frac{1}{2}}$. On the other hand, the smallest singular value of $\mathbf{A}$ will be denoted by $\sigma_{\min}(\mathbf{A})$.

1.2 Basic Facts about (M, g)

Working with Riemannian submersion, we shall as usual start with computations regarding the horizontal and verticle tangent spaces.

Proposition 1.

1. The map Σ defined in (1.5) is a smooth injective immersion.
2. At every $(\mathbf{A}, s) \in \Theta$, the verticle tangent space

$$V_{(\mathbf{A}, s)} := \ker(d\pi_{(\mathbf{A}, s)}) = \ker(d\Phi_{(\mathbf{A}, s)}) \quad (1.7)$$

equals to the collection of all tangent vectors of the form $(\mathbf{A}\mathbf{\Omega}, 0) \in T_{(\mathbf{A}, s)}\Theta$, with $\mathbf{\Omega}$ ranged over all $q \times q$ skew symmetric matrices.

3. At every $(\mathbf{A}, s) \in \Theta$, the horizontal tangent space $H_{(\mathbf{A}, s)} := V_{(\mathbf{A}, s)}^\perp$ equals to the collection of all tangent vectors of the form $(\mathbf{V}, c) \in T_{(\mathbf{A}, s)}\Theta$ ranged over all $c \in \mathbb{R}$ and all $p \times q$ matrices $\mathbf{V}$ satisfying $\mathbf{V}^T \mathbf{A} = \mathbf{A}^T \mathbf{V}$.

Proof. Fix a point $(\mathbf{A}, s) \in \Theta$ throughout this proof. Before we start the proof, let us first study the set $\ker(d\Phi_{(\mathbf{A}, s)})$, where Φ is given in (1.5). For $(\mathbf{V}, c) \in T_{(\mathbf{A}, s)}\Theta$, define a curve $\gamma : (-\epsilon, \epsilon) \rightarrow \Theta$ by setting $\gamma(t) = (\mathbf{A} + t\mathbf{V}, s + tc)$. Then:

$$d\Phi_{(\mathbf{A}, s)}((\mathbf{V}, c)) = \left. \frac{d}{dt} \right|_{t=0} \Phi(\gamma(t)) = \mathbf{A}\mathbf{V}^T + \mathbf{V}\mathbf{A}^T + c\mathbf{I}_p.$$

If $(\mathbf{V}, c) \in \ker(d\Phi_{(\mathbf{A}, s)})$, then

$$\mathbf{A}\mathbf{V}^T + \mathbf{V}\mathbf{A}^T + c\mathbf{I}_p = \mathbf{0}. \quad (1.8)$$

If $\mathbf{u} \in \mathbb{R}^p$ is any nonzero vector orthogonal to all column vectors of $\mathbf{A}$, then $\mathbf{A}^T \mathbf{u} = \mathbf{0}$ and

$$0 = \mathbf{u}^T (\mathbf{A}\mathbf{V}^T + \mathbf{V}\mathbf{A}^T + c\mathbf{I}_p) \mathbf{u} = c \mathbf{u}^T \mathbf{u}.$$

Given that $\mathbf{u} \neq \mathbf{0}$, we must have $c = 0$ and (1.8) reduces to

$$\mathbf{A}\mathbf{V}^T + \mathbf{V}\mathbf{A}^T = \mathbf{0}. \quad (1.9)$$

Apply $\mathbf{u}$ to the right to get:

$$\mathbf{A}\mathbf{V}^T \mathbf{u} = \mathbf{0}.$$

Given that $\mathbf{A}$ has full rank, rank nullity theorem forces

$$\mathbf{V}^T \mathbf{u} = \mathbf{0}.$$

Therefore, we have proven that every vector $\mathbf{u}$ orthogonal to columns vectors of $\mathbf{A}$ will also be orthogonal to column vectors of $\mathbf{V}$. This means $\mathbf{V} = \mathbf{A}\mathbf{\Omega}$ for some $q \times q$ matrix $\mathbf{\Omega}$. Substitute this back to (1.9) gives

$$\mathbf{A}(\mathbf{\Omega}^T + \mathbf{\Omega})\mathbf{A}^T = \mathbf{0}.$$

Applying $(\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T$ to the left and $\mathbf{A}(\mathbf{A}^T \mathbf{A})^{-1}$ to the right yields $\mathbf{\Omega}^T + \mathbf{\Omega} = \mathbf{0}$. Therefore, $\mathbf{\Omega}$ is skew-symmetric. Conversely, (1.9) is satisfied by $\mathbf{V} = \mathbf{A}\mathbf{\Omega}$ with any skew-symmetric matrix $\mathbf{\Omega}$. We have proven that $\ker(d\Phi_{(\mathbf{A},s)})$ equals to the collection of all tangent vectors of the form $(\mathbf{A}\mathbf{\Omega}, 0) \in T_{(\mathbf{A},s)}\Theta$, with $\mathbf{\Omega}$ ranged over all $q \times q$ skew symmetric matrices.

We now move back to prove 1 and 2 at the same time by showing (1.7). Clearly, Σ is well-defined and is injective. The smoothness of Σ is guaranteed by relation $\Sigma \circ \pi = \Phi$ (see for example [Lee(2013)] Theorem 4.29). By the chain rule, we have

$$d\Phi_{(\mathbf{A},s)} = d\Sigma_{[\mathbf{A},s]} \circ d\pi_{(\mathbf{A},s)}. \quad (1.10)$$

It follows immediately that $\ker(d\pi_{(\mathbf{A},s)}) \subseteq \ker(d\Phi_{(\mathbf{A},s)})$. The other direction will follow if we can prove that $d\Sigma_{[\mathbf{A},s]}$ is injective.

Suppose that $\mathbf{v} \in T_{[\mathbf{A},s]}M$ is such that $d\Sigma_{[\mathbf{A},s]}(\mathbf{v}) = 0$. We prove that $\mathbf{v} = \mathbf{0}$. Let $(\mathbf{V}, c) \in T_{(\mathbf{A},s)}\Theta$ be the horizontal lift of $\mathbf{v}$ at $(\mathbf{A}, s)$, meaning that $(\mathbf{V}, c)$ is the unique horizontal tangent vector at $(\mathbf{A}, s)$ such that $\mathbf{v} = d\pi_{(\mathbf{A},s)}((\mathbf{V}, c))$. (1.10) then implies that $(\mathbf{V}, c) \in \ker(d\Phi_{(\mathbf{A},s)})$ and the proceeding discussion shows that $c = 0$ and $\mathbf{V} = \mathbf{A}\mathbf{\Omega}$ for some skew-symmetric $\mathbf{\Omega}$. To proceed, we look at the curve $\gamma : (-\epsilon, \epsilon) \rightarrow \Theta$ defined by $\gamma(t) = (\mathbf{A}e^{t\mathbf{\Omega}}, s)$. Clearly $\gamma(0) = (\mathbf{A}, s)$ and $\gamma'(0) = (\mathbf{A}\mathbf{\Omega}, 0)$. Moreover, the matrix $e^{t\mathbf{\Omega}}$ is always orthogonal for any t and any skew-symmetric $\mathbf{\Omega}$ so that $\pi(\gamma(t)) = [\mathbf{A}, s]$ for all t . We have

$$\mathbf{v} = d\pi_{(\mathbf{A},s)}((\mathbf{A}\mathbf{\Omega}, 0)) = d\pi_{(\mathbf{A},s)}(\gamma'(0)) = \left. \frac{d}{dt} \right|_{t=0} \pi(\gamma(t)) = \mathbf{0} \in T_{[\mathbf{A},s]}M.$$

Therefore, $d\Sigma_{[\mathbf{A},s]}$ is injective and the verticle tangent space is the same as $\ker(d\Phi_{(\mathbf{A},s)})$. Since $[\mathbf{A}, s]$ is arbitrary, Σ is an immersion.

To compute the horizontal tangent space, any $(\mathbf{V}, c) \in V_{(\mathbf{A},s)}^\perp$ will satisfy

$$0 = g_{(\mathbf{A},s)}^\Theta((\mathbf{V}, c), (\mathbf{A}\mathbf{\Omega}, 0)) = \text{tr}(\mathbf{A}^T \mathbf{V} \mathbf{\Omega}) \quad (1.11)$$

for all skew-symmetric $\mathbf{\Omega}$. Decomposing $\mathbf{A}^T \mathbf{V} = \mathbf{S} + \mathbf{K}$, where $\mathbf{S}$ is the symmetric part and $\mathbf{K}$ is the skew-symmetric part, we find out that

$$0 = \text{tr}(\mathbf{S}\mathbf{\Omega}) + \text{tr}(\mathbf{K}\mathbf{\Omega}) = 0 + \text{tr}(\mathbf{K}\mathbf{\Omega}) = \text{tr}(\mathbf{K}\mathbf{\Omega}).$$

Setting $\mathbf{\Omega} = \mathbf{K}^T$ will yield that $\mathbf{K} = \mathbf{0}$. Therefore, $\mathbf{A}^T \mathbf{V}$ is a symmetric matrix. Conversely, any $\mathbf{V}$ such that $\mathbf{A}^T \mathbf{V}$ is symmetric will satisfy (1.11). The proof is complete. $\square$

Proposition 2. Given $[\mathbf{A}, s] \in M$ and $\mathbf{v} \in T_{[\mathbf{A},s]}M$, geodesics in M starting at $[\mathbf{A}, s]$ with initial velocity $\mathbf{v}$ takes the format of curves $\gamma : I \rightarrow M$, where I is an open interval containing 0, defined by $\gamma(t) = [\mathbf{A} + t\mathbf{V}, s + tc]$, where $(\mathbf{V}, c) \in H_{(\mathbf{A},s)}$ is the horizontal lift of $\mathbf{v}$ at $(\mathbf{A}, s)$. Consequently, the exponential map is computed as $\text{Exp}_{[\mathbf{A},s]}(\mathbf{v}) = [\mathbf{A} + \mathbf{V}, s + c]$, at least well-defined over an open ball $\mathcal{E}_{[\mathbf{A},s]}$ of $\mathbf{0} \in T_{[\mathbf{A},s]}M \cong \mathbb{R}^{pq+1-\frac{q(q-1)}{2}}$ of radius $\min(\sigma_{\min}(\mathbf{A}), s)$.

Proof. First of all, let us quickly check that the curve γ is well defined. Indeed, if $(\mathbf{A}\mathbf{Q}, s)$, where $\mathbf{Q} \in O(q)$, is another representative of $[\mathbf{A}, s]$, let $(\mathbf{V}', c')$ be the horizontal lift of $\mathbf{v}$ to $(\mathbf{A}\mathbf{Q}, s)$. With the function $R_{\mathbf{Q}} : \Theta \rightarrow \Theta$ defined by $R_{\mathbf{Q}}((\mathbf{A}, s)) = (\mathbf{A}\mathbf{Q}, s)$, we have

$$\mathbf{v} = d\pi_{(\mathbf{A},s)}(\mathbf{V}, c) = d(\pi \circ R_{\mathbf{Q}})_{(\mathbf{A},s)}(\mathbf{V}, c) = d\pi_{(\mathbf{A}\mathbf{Q},s)}(dR_{\mathbf{Q}})_{(\mathbf{A},s)}(\mathbf{V}, c) = d\pi_{(\mathbf{A}\mathbf{Q},s)}(\mathbf{V}\mathbf{Q}, c).$$

Thus, $\mathbf{V}' = \mathbf{V}\mathbf{Q}$ and $c' = c$. Substitute this into the definition of γ will yield the exact same curve.

A well-known property of geodesics in case of Riemannian submersion (see for example [O'Neill(1967)] Section 2 Corollary 2 or [Valencia(2021)] Theorem 3.6) is that any geodesic c in Θ with initial velocity being horizontal will always have horizontal velocities and $\gamma = \pi \circ c$ will remain a geodesic in M . Therefore, $\text{Exp}_{[\mathbf{A},s]}(\mathbf{v}) = [\mathbf{A} + \mathbf{V}, s + c]$ is well defined for all vectors $\mathbf{v} \in T_{[\mathbf{A},s]}M$ of small enough norm making sure that $(\mathbf{A} + \mathbf{V}, s + c) \in \Theta$. To have a quantified upper bound of the norm, we cite Weyl's perturbation inequality (Chapter 7.3 Problem 16 equation (7.3.15) of [Horn and Johnson(2013)]) to get

$$\sigma_{\min}(\mathbf{A} + \mathbf{V}) \geq \sigma_{\min}(\mathbf{A}) - \|\mathbf{V}\| \geq \sigma_{\min}(\mathbf{A}) - |\mathbf{v}| > 0,$$

whenever $|\mathbf{v}| < \min(\sigma_{\min}(\mathbf{A}), s)$. The last inequality is because the spectral norm of $\mathbf{V}$ is always smaller than the Euclidean norm (Frobenius norm) of $\mathbf{V}$ and hence norm of $\mathbf{v}$. Thus, for all $\mathbf{v} \in \mathcal{E}_{[\mathbf{A},s]}$, $\text{rank}(\mathbf{A} + \mathbf{V}) = q$ and the exponential map is well-defined. $\square$

To compute the logarithmic map, we need first preparatory lemmas that work out an orthogonal matrix based on given points in M .

Lemma 3. If $\mathbf{L}$ is an invertible $q \times q$ matrix and $\mathbf{Q} \in O(q)$ are such that $\mathbf{L}\mathbf{Q}$ is symmetric, then there exists matrices $\mathbf{R} \in O(q)$, $\mathbf{P} \in O(q)$, a $q \times q$ diagonal matrix $\mathbf{D}$ whose diagonal entries are 1 or -1 , and a $q \times q$ diagonal matrix $\mathbf{S}$ containing the singular values of $\mathbf{L}$ in increasing order such that $\mathbf{L} = \mathbf{R}\mathbf{S}\mathbf{P}^T$ and $\mathbf{Q} = \mathbf{P}\mathbf{D}\mathbf{R}^T$.

Proof. Let us start by applying the singular value decomposition of $\mathbf{L}$ to get matrices $\mathbf{R}' \in O(q)$, $\mathbf{P}' \in O(q)$ and $\mathbf{S}$ such that $\mathbf{L} = \mathbf{R}'\mathbf{S}(\mathbf{P}')^T$. Substituting the given condition that $\mathbf{L}\mathbf{Q}$ is symmetric will yield that $\mathbf{S}\mathbf{J} = \mathbf{J}^T\mathbf{S}$, where $\mathbf{J} := (\mathbf{P}')^T\mathbf{Q}\mathbf{R}'$.

Clearly $\mathbf{J} \in O(q)$. The equation

$$\mathbf{S}^2 = \mathbf{S}\mathbf{J}\mathbf{J}^T\mathbf{S} = (\mathbf{S}\mathbf{J})(\mathbf{S}\mathbf{J})^T = (\mathbf{J}^T\mathbf{S})(\mathbf{J}^T\mathbf{S})^T = \mathbf{J}^T\mathbf{S}^2\mathbf{J}$$

shows that

$$\mathbf{J}\mathbf{S}^2 = \mathbf{S}^2\mathbf{J}.$$

Looking at each entry of both size, we end up with

$$\mathbf{J}_{ij}(\mathbf{S}_{jj}^2 - \mathbf{S}_{ii}^2) = 0$$

for $i, j = 1, \dots, q$. Given that $\mathbf{L}$ is invertible, its singular values are all strictly positive. Therefore, in fact we have

$$\mathbf{J}_{ij}(\mathbf{S}_{jj} - \mathbf{S}_{ii}) = 0 \tag{1.12}$$

for all i and all j so that $\mathbf{J}\mathbf{S} = \mathbf{S}\mathbf{J} = \mathbf{J}^T\mathbf{S}$. Invert $\mathbf{S}$ yields that $\mathbf{J}$ is actually symmetric. Being a symmetric orthogonal matrix, $\mathbf{J}^2 = \mathbf{I}_q$ and the eigenvalues of $\mathbf{J}$ must be 1 or -1 . Another look at (1.12) shows that $\mathbf{J}$ is a block-diagonal matrix, diagonal if and only if all diagonal entries of $\mathbf{S}$ are distinct. Furthermore, each block of $\mathbf{J}$ must still be orthogonal and symmetric.

We now diagonalize $\mathbf{J}$ based on the above information. Suppose that $\mathbf{S} = \bigoplus_{i=1}^k \mathbf{s}_i \mathbf{I}_{m_i}$, where $\mathbf{s}_1 < \dots < \mathbf{s}_k$ are the unique diagonal entries of $\mathbf{S}$ and each repeats for m_i times. Thus, $\mathbf{J}$ has k blocks. Write the i th block by $\mathbf{J}_i$ so that $\mathbf{J} = \bigoplus_{i=1}^k \mathbf{J}_i$. The spectral theorem then allows us to orthogonally diagonalize $\mathbf{J}_i = \mathbf{K}_i \mathbf{D}_i \mathbf{K}_i^T$, where $\mathbf{K}_i \in O(m_i)$ and $\mathbf{D}_i$ is an $m_i \times m_i$ diagonal matrix with 1 or -1 diagonal entry. Put $\mathbf{K} = \bigoplus_{i=1}^k \mathbf{K}_i$ and $\mathbf{D} = \bigoplus_{i=1}^k \mathbf{D}_i$. Then $\mathbf{K} \in O(q)$, $\mathbf{D}$ is a $q \times q$ diagonal matrix with 1 or -1 diagonal entry, and $\mathbf{J} = \mathbf{K}\mathbf{D}\mathbf{K}^T$. This involved construction leads us to the outcome that $\mathbf{K}$ in fact commutes with $\mathbf{S}$:

$$\mathbf{K}\mathbf{S} = \bigoplus_{i=1}^k \mathbf{s}_i \mathbf{K}_i = \mathbf{S}\mathbf{K}.$$

Put $P = P'K$, $R = R'K$. We have

$$Q = P'J(R')^T = P'KDK^T(R')^T = PDR^T$$

and

$$RSP^T = R'KSK^T(P')^T = R'SKK^T(P')^T = R'S(P')^T = L.$$

This completes the proof. $\square$

Lemma 4. If L is an invertible $q \times q$ matrix with singular value decomposition $L = RSP^T$, then

$$PR^T = (L^T L)^{-\frac{1}{2}} L^T.$$

This means, although neither P nor R is uniquely determined, PR^T is uniquely determined.

Proof. Given that

$$L^T L = PS^2 P^T,$$

we have

$$(L^T L)^{\frac{1}{2}} = PSP^T.$$

Therefore:

$$L = RP^T PSP^T = RP^T (L^T L)^{\frac{1}{2}}.$$

The result follows from inverting $(L^T L)^{\frac{1}{2}}$ and take transpose. $\square$

Proposition 5. For every $[A, s] \in M$ there exists an open neighborhood $U \subset M$ containing $[A, s]$ such that

1. For every $[B, t] \in U$, $A^T B$ is invertible.
2. The logarithmic map $\text{Exp}_{[A, s]}^{-1} : U \rightarrow T_{[A, s]}M$ is well-defined and is smooth in U . In fact:

$$\text{Exp}_{[A, s]}^{-1}([B, t]) = d\pi_{(A, s)}((BQ(A, B) - A, t - s)), \quad (1.13)$$

where

$$Q(A, B) = (B^T A A^T B)^{-\frac{1}{2}} B^T A \in O(q). \quad (1.14)$$

Proof. First of all, if $R \in O(q)$ and $P \in O(q)$, then $Q(AP, BR) = R^T Q(A, B)P$ and the same computation as in the proof of Proposition 2 shows that (1.13) is indeed well-defined.

It is a well-known fact that $\text{Exp}_{[A, s]}$ is a local diffeomorphism and thus there exists an open neighborhood containing $[A, s]$ in which the logarithmic map $\text{Exp}_{[A, s]}^{-1}$ is well-defined and is smooth. Given that $\text{Exp}_{[A, s]}^{-1}([A, s]) = 0$, we shall only take an open subset U_1 of this open neighborhood such that, whenever $[B, t] \in U_1$ and $v = \text{Exp}_{[A, s]}^{-1}([B, t])$, we always have $|v| < \epsilon$, where

$$\epsilon \in \left(0, \min \left(\sigma_{\min}(A), 2 \left[\frac{2}{\sigma_{\min}(A)} + 1 \right]^{-1} \right) \right)$$

is a fixed constant.

Next, let $U^\Theta \subset \Theta$ be the collection of all $(B, t) \in \Theta$ such that $\|A - B\| < \epsilon$. Then U^Θ is an open neighborhood of (A, s) and any $(B, t) \in U^\Theta$ has the property that $A^T B$ is invertible thanks to the condition that $\epsilon < \sigma_{\min}(A)$. To see this, suppose for the sake of contradiction that there exists a unit vector $x \in \mathbb{R}^q$ with $A^T Bx = 0$. Then

$$|Ax|^2 = x^T A^T A x = -x^T A^T (B - A)x \leq |Ax| \|B - A\| \cdot 1$$

and $\|B - A\| \geq \|Ax\| \geq \sigma_{\min}(A)$, a contradiction. The fact that $A^T B$ is invertible guarantees that the matrix $Q(A, B)$ given at (1.14) is well defined for any $(B, t) \in U_\Theta$ and, by continuity of $Q(A, B)$ and the fact that $Q(A, A) = I_q$, we can shrink U^Θ to also guarantee that $\|Q(A, B) - I_q\| < \epsilon$ whenever $(B, t) \in U_\Theta$. We define $U_2 := \pi(U^\Theta)$. Given that π is an open map, U_2 is an open neighborhood of $[A, s]$. If $[B, t] \in U_2$, then *every* representative (B, t) of $[B, t]$ will have $A^T B$ being invertible because $\det(A^T B R) = \det(A^T B) \neq 0$ for every $R \in O(q)$, although only *some* representatives (B, t) will satisfy both $\|A - B\| < \epsilon$ and $\|Q(A, B) - I_q\| < \epsilon$. Given that our results do not depend on representatives, we shall always choose a representative (B, t) such that this is the case.

We now compute $\text{Exp}_{[A, s]}^{-1}$ for $[B, t] \in U := U_1 \cap U_2$. By Proposition 2, $[B, t] = [A + V, s + c]$, where (V, c) is the horizontal lift of $v = \text{Exp}_{[A, s]}^{-1}([B, t])$ to (A, s) . It follows that $c = t - s$ and that there exists a $Q \in O(q)$ such that $V = BQ - A$. Or equivalently:

$$\text{Exp}_{[A, s]}^{-1}([B, t]) = d\pi_{(A, s)}((BQ - A, t - s)). \quad (1.15)$$

Given that (V, c) is a horizontal tangent vector, Proposition 1 shows that

$$A^T(BQ - A) = (BQ - A)^T A$$

and consequently $A^T BQ$ is symmetric. Lemma 3 with $L = A^T B$ shows the existence of $R, P \in O(q)$, a diagonal matrix S with the singular values of $A^T B$ in increasing order, and a $q \times q$ diagonal matrix D with diagonal entries 1 or -1 such that $A^T B = RSP^T$ and $Q = PDR^T$. (1.15) now becomes

$$\text{Exp}_{[A, s]}^{-1}([B, t]) = d\pi_{(A, s)}((BPDR^T - A, t - s)). \quad (1.16)$$

Now, given that Lemma 4 shows that $Q(A, B) = PR^T$, the only thing remaining is to rule out all $D \neq I_q$. We achieve this by comparing the two matrices directly. We have

$$\|Q - Q(A, B)\| = \|QQ(A, B)^T - I_q\| = \|PDP^T - I_q\|.$$

Given that

$$(PDP^T - I_q)^2 = P(D^2 - 2D + I_q)P^T,$$

we see that $\|Q - Q(A, B)\|$ equals to either 0 or 2 and takes no other values. It suffices to bound it away from 2. We have

$$\begin{aligned} \|QQ(A, B)^T - I_q\| &\leq \|(Q - I_q)Q(A, B)^T + (Q(A, B)^T - I_q)\| \leq \|Q(A, B)^T - I_q\| + \|Q - I_q\| \\ &< \epsilon + \frac{1}{\sigma_{\min}(A)} \|AQ - A\| \leq \epsilon + \frac{1}{\sigma_{\min}(A)} \|AQ - BQ + V\| \leq \epsilon + \frac{1}{\sigma_{\min}(A)} (\|A - B\| + \|V\|) \\ &\leq \epsilon + \frac{2\epsilon}{\sigma_{\min}(A)} < 2. \end{aligned}$$

This completes the proof. $\square$

Remark 6. Unfortunately, the logarithmic map is not globally defined in the standard way even over all $[B, t]$ such that $A^T B$ is invertible. For a counterexample, take $p = 2, q = 1, A = [1, 0]^T, B = [1, 1]^T$ and $s = t = 1$. In this case, $O(1) = \{1, -1\}$. Then, although $A^T B = 1$ is invertible, we still have

$$\text{Exp}_{[A, s]}(d\pi_{(A, s)}((B - A, 0))) = [B, 1] = [-B, 1] = \text{Exp}_{[A, s]}(d\pi_{(A, s)}((-B - A, 0))).$$

The reason is that, even when we assume $A^T B$ is invertible, deducing $D = I_q$ relies on the local smoothness of the logarithmic map and the procedure of letting $[B, t] \rightarrow [A, s]$. If $[B, t]$ is far away from $[A, s]$, then $Q = PDR^T$ with some $D \neq I_q$ can still produce a valid horizontal vector whose image under $\text{Exp}_{[A, s]}$ is $[B, t]$, so that $\text{Exp}_{[A, s]}$ fails to be injective and admits no globally defined inverse.

Nevertheless, the choice $D = I_q$ has an interesting property of selecting the geodesic of minimal speed and length when $[B, t]$ is any point such that $A^T B$ is invertible. To see this, consider a general candidate $Q = PDR^T$ and suppose that $v = d\pi_{(A,s)}((BQ - A, t - s))$ satisfies $\text{Exp}_{[A,s]}(v) = [B, t]$. Given that so that $(BQ - A, t - s)$ is the horizontal lift of v at (A, s) , we can compute the squared norm of v :

$$\begin{aligned} g_{[A,s]}(v, v) &= g_{(A,s)}^\Theta((BQ - A, t - s), (BQ - A, t - s)) \\ &= \text{tr}((BQ - A)(BQ - A)^T) + (t - s)^2 = \text{tr}(BB^T) + \text{tr}(AA^T) - 2\text{tr}(SD) + (t - s)^2, \end{aligned}$$

where the last equality substituted $A^T BQ = RSP^T PDR^T = RSDR^T$ and used the cyclic property of the trace. The fact that the diagonal entries of S are strictly positive show that $g_{[A,s]}(v, v)$ is minimized precisely when $D = I_q$. Given that the geodesic starting at $[A, s]$ with initial velocity v reaches $[B, t]$ at time 1 with constant speed $\sqrt{g_{[A,s]}(v, v)}$, its length is $\sqrt{g_{[A,s]}(v, v)}$. The choice $D = I_q$ gives the slowest and hence shortest geodesic. To illustrate, let us go back to the counterexample in the previous paragraph. Put $v_\pm := d\pi_{(A,s)}((\pm B - A, 0))$. Direct computation shows $g_{[A,s]}(v_+, v_+) = 1$ and $g_{[A,s]}(v_-, v_-) = 5$ so that the option corresponding to $D = I_1 = 1$ gives the slowest speed. Therefore, it is actually possible to define $\text{Exp}_{[A,s]}^{-1}$ over all $[B, t]$ such that $A^T B$ is invertible by manually asking it to output the minimal speed candidate and this definition will yield the same $Q(A, B)$ formula.

However, the situation totally breaks down in case $A^T B$ is not invertible. Consider again $p = 2, q = 1$, $A = [1, 0]^T, s = t = 1$, but this time $B = [0, 1]^T$. Then $g_{[A,s]}(v_+, v_+) = g_{[A,s]}(v_-, v_-) = 2$ and there are two distinct geodesics of the same speed so $\text{Exp}_{[A,s]}^{-1}([B, t])$ cannot be well-defined.

(Picture: there is a hole at $[0, 0]^T$. Starting $A = [1, 0]^T$, as $B = [0, 1]^T$ is glued to $[0, -1]^T$, v_+ points from $[1, 0]^T$ to $[0, 1]^T$, v_- points from $[1, 0]^T$ to $[0, -1]^T$)

Given the above situation and (the expectation that) our work should not require a global definition, we stick to the standard definition of logarithmic map only defined locally around $[A, s]$.

1.3 Properties of the Probabilistic Principal Component Analysis Model

We now show that the statistical model $\mathcal{P}$ given by (1.6) is *differentiable in quadratic mean* and has *positive definite Fisher information operator* in the sense developed in [Sun et al.(2026)Sun, Lin, and Liu], definition III.1 and III.2.

Let us introduce a few notations. For each $[A, s] \in M$, let

$$p(x; [A, s]) = (2\pi)^{-\frac{p}{2}} \det^{-\frac{1}{2}}(\Sigma([A, s])) \exp\left(-\frac{1}{2}x^T(\Sigma([A, s]))^{-1}x\right) \quad (1.17)$$

be the density of $N_p(0, \Sigma([A, s]))$. For $v \in \mathcal{E}_{[A,s]} \subset T_{[A,s]}M \cong \mathbb{R}^{pq+1-\frac{q(q-1)}{2}}$, write $l_{[A,s]}(v)$ for the log-likelihood function of the local experiment

$$l_{[A,s]}(v)(x) = \log\left(p(x; \text{Exp}_{[A,s]}(v))\right). \quad (1.18)$$

Define the square root likelihood function $r_{[A,s]} : \mathcal{E}_{[A,s]} \rightarrow L^2(\mathbb{R}^p)$ by setting, for all $v \in \mathcal{E}_{[A,s]}$, that

$$r_{[A,s]}(v)(x) = \sqrt{p(x; \text{Exp}_{[A,s]}(v))}. \quad (1.19)$$

Theorem 7. $r_{[A,s]}$ is Fréchet differentiable at any $v \in \mathcal{E}_{[A,s]}$. In particular, $\mathcal{P}$ is differentiable in quadratic mean with score

$$s(x; [A, s])(h) = \frac{1}{2} \text{tr} \left\{ (\Sigma([A, s]))^{-1} [xx^T - (\Sigma([A, s]))] (\Sigma([A, s]))^{-1} (D_h \Sigma([A, s])) \right\}, \quad (1.20)$$

where

$$D_{\mathbf{h}}\Sigma([\mathbf{A}, s]) := \left. \frac{d}{dt} \right|_{t=0} \Sigma(\text{Exp}_{[\mathbf{A}, s]}(t\mathbf{h})) = \mathbf{U}\mathbf{A}^T + \mathbf{A}\mathbf{U}^T + b\mathbf{I}_p, \quad (1.21)$$

if $\mathbf{h} \in T_{[\mathbf{A}, s]}M$ lifts to $(\mathbf{U}, b)$ at $(\mathbf{A}, s)$. Furthermore, $\mathcal{P}$ is a regular model in the sense that the Fisher information operator

$$G_{[\mathbf{A}, s]}(\mathbf{h}_1, \mathbf{h}_2) = \frac{1}{2} \text{tr} \left\{ (\Sigma([\mathbf{A}, s]))^{-1} D_{\mathbf{h}_1}\Sigma([\mathbf{A}, s]) (\Sigma([\mathbf{A}, s]))^{-1} D_{\mathbf{h}_2}\Sigma([\mathbf{A}, s]) \right\} \quad (1.22)$$

is positive definite.

Proof. The proof for the Fréchet differentiability is long but routine. We compute the Gâteaux derivative of $r_{[\mathbf{A}, s]}$ at $\mathbf{v} \in \mathcal{E}_{[\mathbf{A}, s]}$ in a given direction $\mathbf{h} \in T_{[\mathbf{A}, s]}M$ and show that the Gâteaux derivative is the Fréchet derivative by a direct analysis of the remainder term. At the very end, we specialize our computations to $\mathbf{v} = \mathbf{0}$ to prove (1.20) and (1.22).

Given $\mathbf{v}$ and $\mathbf{h}$, there exists an $\epsilon > 0$ such that $\mathbf{v} + t\mathbf{h} \in \mathcal{E}_{[\mathbf{A}, s]}$ whenever $t \in (-\epsilon, \epsilon)$. If they respectively lift to $(\mathbf{V}, c)$ and $(\mathbf{U}, b)$ at $(\mathbf{A}, s)$, then $\mathbf{v} + t\mathbf{h}$ lifts to $(\mathbf{V} + t\mathbf{U}, c + tb)$ at $(\mathbf{A}, s)$. First of all:

$$\begin{aligned} D_{\mathbf{h}}\Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) &:= \left. \frac{d}{dt} \right|_{t=0} \Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v} + t\mathbf{h})) \\ &= \left. \frac{d}{dt} \right|_{t=0} \left\{ (\mathbf{A} + \mathbf{V} + t\mathbf{U})(\mathbf{A} + \mathbf{V} + t\mathbf{U})^T + (s + c + tb)\mathbf{I}_p \right\} \\ &= \mathbf{U}\mathbf{V}^T + \mathbf{V}\mathbf{U}^T + \mathbf{U}\mathbf{A}^T + \mathbf{A}\mathbf{U}^T + b\mathbf{I}_p, \end{aligned} \quad (1.23)$$

proving (1.21) with special case $\mathbf{v} = \mathbf{0}$. Secondly, for all $\mathbf{x} \in \mathbb{R}^p$:

$$\begin{aligned} D_{\mathbf{h}}l_{[\mathbf{A}, s]}(\mathbf{v})(\mathbf{x}) &:= \left. \frac{d}{dt} \right|_{t=0} l_{[\mathbf{A}, s]}(\mathbf{v} + t\mathbf{h})(\mathbf{x}) \\ &= \left. \frac{d}{dt} \right|_{t=0} \left\{ -\frac{p}{2} \log(2\pi) - \frac{1}{2} \log \det \left(\Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v} + t\mathbf{h})) \right) - \frac{1}{2} \mathbf{x}^T \left(\Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v} + t\mathbf{h})) \right)^{-1} \mathbf{x} \right\} \\ &= -\frac{1}{2} \text{tr} \left[\left(\Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right)^{-1} \left(\left. \frac{d}{dt} \right|_{t=0} \Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v} + t\mathbf{h})) \right) \right] \\ &\quad + \frac{1}{2} \mathbf{x}^T \left(\Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right)^{-1} \left(\left. \frac{d}{dt} \right|_{t=0} \Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v} + t\mathbf{h})) \right) \left(\Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right)^{-1} \mathbf{x} \\ &= -\frac{1}{2} \text{tr} \left[\left(\Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right)^{-1} \left(D_{\mathbf{h}}\Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right) \right] \\ &\quad + \frac{1}{2} \mathbf{x}^T \left(\Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right)^{-1} \left(D_{\mathbf{h}}\Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right) \left(\Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right)^{-1} \mathbf{x} \\ &= \frac{1}{2} \text{tr} \left\{ \left(\Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right)^{-1} \left[\mathbf{x}\mathbf{x}^T - \left(\Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right) \right] \left(\Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right)^{-1} \left(D_{\mathbf{h}}\Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right) \right\}. \end{aligned} \quad (1.24)$$

Finally:

$$\begin{aligned} D_{\mathbf{h}}r_{[\mathbf{A}, s]}(\mathbf{v})(\mathbf{x}) &= D_{\mathbf{h}} \exp \left(\frac{1}{2} l_{[\mathbf{A}, s]}(\mathbf{v})(\mathbf{x}) \right) = \frac{1}{2} \exp \left(\frac{1}{2} l_{[\mathbf{A}, s]}(\mathbf{v})(\mathbf{x}) \right) D_{\mathbf{h}}l_{[\mathbf{A}, s]}(\mathbf{v})(\mathbf{x}) \\ &= \frac{1}{2} r_{[\mathbf{A}, s]}(\mathbf{v})(\mathbf{x}) D_{\mathbf{h}}l_{[\mathbf{A}, s]}(\mathbf{v})(\mathbf{x}). \end{aligned} \quad (1.25)$$

To prove that (1.25) indeed defines a valid Gâteaux derivative, we must prove that $D_{\mathbf{h}}r_{[\mathbf{A}, s]}(\mathbf{v}) \in L^2(\mathbb{R}^p)$. Luckily, the expectation and covariance of normal quadratic forms have already been very well studied in [Gupta and Nagar(2000)]. For $\mathbf{X} \sim N_p(\mathbf{0}, \Sigma)$, Theorem 7.7.1(i) states that

$$\mathbb{E} [\mathbf{X}^T \mathbf{Z} \mathbf{X}] = \text{tr} (\mathbf{Z} \Sigma)$$

for any $p \times p$ matrix $\mathbf{Z}$ and Theorem 7.7.2 states that

$$\text{Cov}(\mathbf{X}^T \mathbf{Z} \mathbf{X}, \mathbf{X}^T \mathbf{Y} \mathbf{X}) = \text{tr}(\mathbf{Z} \Sigma \mathbf{Y}^T \Sigma) + \text{tr}(\mathbf{Z}^T \Sigma \mathbf{Y}^T \Sigma)$$

for any two $p \times p$ matrices $\mathbf{Y}$ and $\mathbf{Z}$. Therefore, we can confirm without effort that the expectation of derivative of log density is zero:

$$\begin{aligned} & \int_{\mathbb{R}^p} D_{\mathbf{h}} l_{[\mathbf{A}, s]}(\mathbf{v})(\mathbf{x}) p(\mathbf{x}; \Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v}))) d\mathbf{x} \\ &= -\frac{1}{2} \text{tr} \left[\left(\Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right)^{-1} \left(D_{\mathbf{h}} \Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right) \right] \\ &+ \frac{1}{2} \int_{\mathbb{R}^p} \left[\mathbf{x}^T \left(\Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right)^{-1} \left(D_{\mathbf{h}} \Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right) \left(\Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right)^{-1} \mathbf{x} \right] p(\mathbf{x}; \Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v}))) d\mathbf{x} = 0 \end{aligned}$$

and, whenever $\mathbf{h}_1$ and $\mathbf{h}_2 \in T_{[\mathbf{A}, s]}M$:

$$\begin{aligned} & \int_{\mathbb{R}^p} D_{\mathbf{h}_1} l_{[\mathbf{A}, s]}(\mathbf{v})(\mathbf{x}) D_{\mathbf{h}_2} l_{[\mathbf{A}, s]}(\mathbf{v})(\mathbf{x}) p(\mathbf{x}; \Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v}))) d\mathbf{x} \\ &= \frac{1}{4} \text{Cov}(\mathbf{X}^T \mathbf{Z}_1 \mathbf{X}, \mathbf{X}^T \mathbf{Z}_2 \mathbf{X}) = \frac{1}{4} \cdot 2 \text{tr} \left(\mathbf{Z}_1 \Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \mathbf{Z}_2 \Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right) \\ &= \frac{1}{2} \text{tr} \left\{ \left(\Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right)^{-1} D_{\mathbf{h}_1} \Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \left(\Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right)^{-1} D_{\mathbf{h}_2} \Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right\}, \end{aligned} \quad (1.26)$$

where $\mathbf{X} \sim N_p(\mathbf{0}, \Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})))$ and

$$\mathbf{Z}_j := \left(\Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right)^{-1} \left(D_{\mathbf{h}_j} \Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right) \left(\Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right)^{-1} \quad (1.27)$$

is symmetric for $j = 1, 2$, so that the two terms in Theorem 7.7.2 of [Gupta and Nagar(2000)] coincide. Thus, in the particular case $\mathbf{h} = \mathbf{h}_1 = \mathbf{h}_2$:

$$\begin{aligned} & \|D_{\mathbf{h}} r_{[\mathbf{A}, s]}(\mathbf{v}); L^2(\mathbb{R}^p)\|^2 = \int_{\mathbb{R}^p} [D_{\mathbf{h}} r_{[\mathbf{A}, s]}(\mathbf{v})(\mathbf{x})]^2 d\mathbf{x} \\ &= \frac{1}{4} \int_{\mathbb{R}^p} (D_{\mathbf{h}} l_{[\mathbf{A}, s]}(\mathbf{v})(\mathbf{x}))^2 p(\mathbf{x}; \Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v}))) d\mathbf{x} = \frac{1}{8} \text{tr} \left\{ \left[\left(\Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right)^{-1} D_{\mathbf{h}} \Sigma(\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v})) \right]^2 \right\}. \end{aligned} \quad (1.28)$$

Therefore, $D_{\mathbf{h}} r_{[\mathbf{A}, s]}(\mathbf{v}) \in L^2(\mathbb{R}^p)$ and (1.25) is indeed a valid Gâteaux derivative at $\mathbf{v}$ in the direction $\mathbf{h}$.

To prove that this is indeed the Fréchet derivative, let us look at the remainder term:

$$R(\mathbf{v}, \mathbf{h})(\mathbf{x}) := r_{[\mathbf{A}, s]}(\mathbf{v} + \mathbf{h})(\mathbf{x}) - r_{[\mathbf{A}, s]}(\mathbf{v})(\mathbf{x}) - D_{\mathbf{h}} r_{[\mathbf{A}, s]}(\mathbf{v})(\mathbf{x})$$

and prove that

$$\lim_{\mathbf{h} \rightarrow \mathbf{0}} \frac{1}{|\mathbf{h}|^2} \|R(\mathbf{v}, \mathbf{h}); L^2(\mathbb{R}^p)\|^2 = 0.$$

To start, the fundamental theorem of calculus shows that

$$\begin{aligned} & r_{[\mathbf{A}, s]}(\mathbf{v} + \mathbf{h})(\mathbf{x}) - r_{[\mathbf{A}, s]}(\mathbf{v})(\mathbf{x}) = \int_0^1 \frac{d}{dt} r_{[\mathbf{A}, s]}(\mathbf{v} + t\mathbf{h})(\mathbf{x}) dt \\ &= \int_0^1 \lim_{s \rightarrow 0} \frac{r_{[\mathbf{A}, s]}(\mathbf{v} + (t+s)\mathbf{h})(\mathbf{x}) - r_{[\mathbf{A}, s]}(\mathbf{v} + t\mathbf{h})(\mathbf{x})}{s} dt = \int_0^1 D_{\mathbf{h}} r_{[\mathbf{A}, s]}(\mathbf{v} + t\mathbf{h})(\mathbf{x}) dt. \end{aligned}$$

Therefore, by Minkowski's integral inequality and the linearity of $D_{\mathbf{h}}r_{[\mathbf{A},s]}(\mathbf{v})$ in $\mathbf{h}$, we have

$$\begin{aligned} \|R(\mathbf{v}, \mathbf{h}); L^2(\mathbb{R}^p)\| &= \left\| \int_0^1 (D_{\mathbf{h}}r_{[\mathbf{A},s]}(\mathbf{v} + t\mathbf{h})(\mathbf{x}) - D_{\mathbf{h}}r_{[\mathbf{A},s]}(\mathbf{v})(\mathbf{x})) dt; L^2(\mathbb{R}^p) \right\| \\ &\leq \int_0^1 \|D_{\mathbf{h}}r_{[\mathbf{A},s]}(\mathbf{v} + t\mathbf{h}) - D_{\mathbf{h}}r_{[\mathbf{A},s]}(\mathbf{v}); L^2(\mathbb{R}^p)\| dt = |\mathbf{h}| \int_0^1 \|D_{\frac{\mathbf{h}}{|\mathbf{h}|}}r_{[\mathbf{A},s]}(\mathbf{v} + t\mathbf{h}) - D_{\frac{\mathbf{h}}{|\mathbf{h}|}}r_{[\mathbf{A},s]}(\mathbf{v}); L^2(\mathbb{R}^p)\| dt \end{aligned}$$

and

$$\begin{aligned} \frac{1}{|\mathbf{h}|^2} \|R(\mathbf{v}, \mathbf{h}); L^2(\mathbb{R}^p)\|^2 &\leq \left\{ \int_0^1 \|D_{\frac{\mathbf{h}}{|\mathbf{h}|}}r_{[\mathbf{A},s]}(\mathbf{v} + t\mathbf{h}) - D_{\frac{\mathbf{h}}{|\mathbf{h}|}}r_{[\mathbf{A},s]}(\mathbf{v}); L^2(\mathbb{R}^p)\| dt \right\}^2 \\ &\leq \left\{ \int_0^1 \sup_{|\mathbf{u}|=1} \|D_{\mathbf{u}}r_{[\mathbf{A},s]}(\mathbf{v} + t\mathbf{h}) - D_{\mathbf{u}}r_{[\mathbf{A},s]}(\mathbf{v}); L^2(\mathbb{R}^p)\| dt \right\}^2. \end{aligned}$$

The proof will be complete once we can show that

$$\lim_{\mathbf{h} \rightarrow \mathbf{0}} \int_0^1 \sup_{|\mathbf{u}|=1} \|D_{\mathbf{u}}r_{[\mathbf{A},s]}(\mathbf{v} + t\mathbf{h}) - D_{\mathbf{u}}r_{[\mathbf{A},s]}(\mathbf{v}); L^2(\mathbb{R}^p)\| dt = 0. \quad (1.29)$$

We must therefore examine the continuity of $D_{\mathbf{h}}r_{[\mathbf{A},s]}(\mathbf{v})$, which is easy to see if we revisit the formulas. (1.23) shows that $D_{\mathbf{h}}\Sigma(\text{Exp}_{[\mathbf{A},s]}(\mathbf{v}))$ is smooth in both $\mathbf{h}$ and $\mathbf{v}$. Thus, (1.25) shows that, for each fixed $\mathbf{x} \in \mathbb{R}^p$, $D_{\mathbf{h}}r_{[\mathbf{A},s]}(\mathbf{v})(\mathbf{x})$ is smooth in both $\mathbf{h}$ and $\mathbf{v}$ and (1.28) shows that $\|D_{\mathbf{h}}r_{[\mathbf{A},s]}(\mathbf{v}); L^2(\mathbb{R}^p)\|$ is smooth in both $\mathbf{h}$ and $\mathbf{v}$. Scheffé's lemma shows that $D_{\mathbf{h}}r_{[\mathbf{A},s]}(\mathbf{v})$, as a map to $L^2(\mathbb{R}^p)$, is continuous in $\mathbf{h}$ and in $\mathbf{v}$. The integrand inside (1.29) is thus bounded uniformly over all $t \in [0, 1]$ and all $\mathbf{h}$ in a compact neighborhood of $\mathbf{0}$ and it suffices to show that, for every $t \in [0, 1]$, that

$$\lim_{\mathbf{h} \rightarrow \mathbf{0}} \sup_{|\mathbf{u}|=1} \|D_{\mathbf{u}}r_{[\mathbf{A},s]}(\mathbf{v} + t\mathbf{h}) - D_{\mathbf{u}}r_{[\mathbf{A},s]}(\mathbf{v}); L^2(\mathbb{R}^p)\| = 0, \quad (1.30)$$

thanks to the bounded convergence theorem. To prove (1.30), let $\mathbf{e}_1, \dots, \mathbf{e}_d$ be a $g_{[\mathbf{A},s]}$ orthonormal basis where $d = pq + 1 - \frac{q(q-1)}{2}$. Every unit vector $\mathbf{u}$ can be expressed as $\mathbf{u} = \sum_{i=1}^d u_i \mathbf{e}_i$ with $\sum_{i=1}^d u_i^2 = 1$. By linearity of $D_{\mathbf{h}}r_{[\mathbf{A},s]}(\mathbf{v})$ in $\mathbf{h}$, we have

$$\begin{aligned} \|D_{\mathbf{u}}r_{[\mathbf{A},s]}(\mathbf{v} + t\mathbf{h}) - D_{\mathbf{u}}r_{[\mathbf{A},s]}(\mathbf{v}); L^2(\mathbb{R}^p)\| &\leq \sum_{i=1}^d |u_i| \|D_{\mathbf{e}_i}r_{[\mathbf{A},s]}(\mathbf{v} + t\mathbf{h}) - D_{\mathbf{e}_i}r_{[\mathbf{A},s]}(\mathbf{v}); L^2(\mathbb{R}^p)\| \\ &\leq \sqrt{d} \max_{i=1, \dots, d} \|D_{\mathbf{e}_i}r_{[\mathbf{A},s]}(\mathbf{v} + t\mathbf{h}) - D_{\mathbf{e}_i}r_{[\mathbf{A},s]}(\mathbf{v}); L^2(\mathbb{R}^p)\|. \end{aligned}$$

Taking supremum yields

$$\sup_{|\mathbf{u}|=1} \|D_{\mathbf{u}}r_{[\mathbf{A},s]}(\mathbf{v} + t\mathbf{h}) - D_{\mathbf{u}}r_{[\mathbf{A},s]}(\mathbf{v}); L^2(\mathbb{R}^p)\| \leq \sqrt{d} \max_{i=1, \dots, d} \|D_{\mathbf{e}_i}r_{[\mathbf{A},s]}(\mathbf{v} + t\mathbf{h}) - D_{\mathbf{e}_i}r_{[\mathbf{A},s]}(\mathbf{v}); L^2(\mathbb{R}^p)\|.$$

Given that $D_{\mathbf{e}_i}r_{[\mathbf{A},s]}(\mathbf{v} + t\mathbf{h}) \rightarrow D_{\mathbf{e}_i}r_{[\mathbf{A},s]}(\mathbf{v})$ in $L^2(\mathbb{R}^p)$ as $\mathbf{h} \rightarrow \mathbf{0}$ for every i , every t , and every $\mathbf{v}$, the proof of Fréchet differentiability is complete.

Finally, the formulas (1.23), (1.24), and (1.25) simplifies greatly when $\mathbf{v} = \mathbf{0}$:

$$\begin{aligned} D_{\mathbf{h}}r_{[\mathbf{A},s]}(\mathbf{0})(\mathbf{x}) &= \frac{1}{2}r_{[\mathbf{A},s]}(\mathbf{0})(\mathbf{x})D_{\mathbf{h}}l_{[\mathbf{A},s]}(\mathbf{0})(\mathbf{x}) \\ &= \frac{1}{4}\sqrt{p(\mathbf{x}; [\mathbf{A}, s])} \text{tr} \left\{ (\Sigma([\mathbf{A}, s]))^{-1} [\mathbf{x}\mathbf{x}^T - (\Sigma([\mathbf{A}, s]))] (\Sigma([\mathbf{A}, s]))^{-1} (D_{\mathbf{h}}\Sigma([\mathbf{A}, s])) \right\}. \end{aligned}$$

The score $s(\mathbf{x}; [\mathbf{A}, s])(\mathbf{h})$ therefore equals to $D_{\mathbf{h}}l_{[\mathbf{A},s]}(\mathbf{0})(\mathbf{x})$ and is indeed given by (1.20). The Fisher information operator (1.22) is obtained by taking $\mathbf{v} = \mathbf{0}$ in (1.26). To show that $G_{[\mathbf{A},s]}$ is positive definite, let $\mathbf{h} \in T_{[\mathbf{A},s]}M$ be

given. We show that $G_{[\mathbf{A}, s]}(\mathbf{h}, \mathbf{h}) \geq 0$ and equality holds only when $\mathbf{h} = \mathbf{0}$. The non-negative part is clear from (1.22) (1.22):

$$G_{[\mathbf{A}, s]}(\mathbf{h}, \mathbf{h}) = \frac{1}{2} \operatorname{tr} \left\{ [(\Sigma([\mathbf{A}, s]))^{-1} D_{\mathbf{h}} \Sigma([\mathbf{A}, s])]^2 \right\} \geq 0.$$

Assume from now on that the equality holds. A slight annoyance here is that the product of two symmetric matrices is not necessarily symmetric and we cannot immediately conclude that $D_{\mathbf{h}} \Sigma([\mathbf{A}, s]) = \mathbf{0}$. We easily overcome this difficulty by using the cyclic property of the trace to get a symmetric matrix

$$\mathbf{N} := (\Sigma([\mathbf{A}, s]))^{-\frac{1}{2}} D_{\mathbf{h}} \Sigma([\mathbf{A}, s]) (\Sigma([\mathbf{A}, s]))^{-\frac{1}{2}}$$

such that

$$0 = G_{[\mathbf{A}, s]}(\mathbf{h}, \mathbf{h}) = \frac{1}{2} \operatorname{tr}(\mathbf{N}^2).$$

Therefore, $\mathbf{N} = \mathbf{0}$ and $D_{\mathbf{h}} \Sigma([\mathbf{A}, s]) = \mathbf{0}$. Now (1.21) reads the same as (1.8) in the proof of Proposition 1. We therefore get $(\mathbf{U}, b) \in \ker(d\Phi_{(\mathbf{A}, s)}) = V_{(\mathbf{A}, s)}$. However, being lifted from $\mathbf{h}$, $(\mathbf{U}, b)$ must also be a horizontal vector. Therefore, $\mathbf{h} = \mathbf{0}$. We have proven that $G_{[\mathbf{A}, s]}$ is positive definite. $\square$

Proposition 8. Suppose that $[\mathbf{A}, s] \in M$ is the true parameter so that each $\mathbf{X}_i$ has density $p(\cdot; [\mathbf{A}, s])$ as in (1.17). We have the local asymptotic normality expansion

$$\sum_{i=1}^n \log \frac{p\left(\mathbf{X}_i; \operatorname{Exp}_{[\mathbf{A}, s]} \left(\frac{\mathbf{h}}{\sqrt{n}} \right)\right)}{p(\mathbf{X}_i; [\mathbf{A}, s])} = \sum_{i=1}^n s(\mathbf{X}_i; [\mathbf{A}, s]) \left(\frac{\mathbf{h}}{\sqrt{n}} \right) - \frac{1}{2} G_{[\mathbf{A}, s]}(\mathbf{h}, \mathbf{h}) + o_{\mathbb{P}}(1) \quad (1.31)$$

whenever n large enough that $\frac{\mathbf{h}}{\sqrt{n}} \in \mathcal{E}_{[\mathbf{A}, s]}$. In particular,

$$\sum_{i=1}^n \log \frac{p\left(\mathbf{X}_i; \operatorname{Exp}_{[\mathbf{A}, s]} \left(\frac{\mathbf{h}}{\sqrt{n}} \right)\right)}{p(\mathbf{X}_i; [\mathbf{A}, s])} \rightarrow N \left(-\frac{1}{2} G_{[\mathbf{A}, s]}(\mathbf{h}, \mathbf{h}), G_{[\mathbf{A}, s]}(\mathbf{h}, \mathbf{h}) \right) \quad (1.32)$$

in distribution as $n \rightarrow \infty$.

Proof. Given that $\mathcal{P}$ is differentiable in quadratic mean and has positive definite Fisher operator, this follows from Proposition III.1 in [Sun et al.(2026)Sun, Lin, and Liu]. $\square$

1.4 Properties of the Maximum Likelihood Estimator of $\mathcal{P}$

We now move to the core of the subject. Throughout this section we fix a *true parameter* $[\mathbf{A}_0, s_0] \in M$, we assume that $\mathbf{X}_1, \mathbf{X}_2, \dots$ are independent and identically distributed random vectors with common density $p(\cdot; [\mathbf{A}_0, s_0])$ as in (1.17), and we abbreviate

$$\Sigma_0 := \Sigma([\mathbf{A}_0, s_0]) = \mathbf{A}_0 \mathbf{A}_0^T + s_0 \mathbf{I}_p, \quad \mathbf{S}_n := \frac{1}{n} \sum_{i=1}^n \mathbf{X}_i \mathbf{X}_i^T, \quad d := pq + 1 - \frac{q(q-1)}{2} = \dim M. \quad (1.33)$$

We write $\rightsquigarrow$ for convergence in distribution and, following (1.4), we abbreviate $|\mathbf{v}| := \sqrt{g_{[\mathbf{A}, s]}(\mathbf{v}, \mathbf{v})}$ for $\mathbf{v} \in T_{[\mathbf{A}, s]} M$.

The whole point of the construction of Section 1.1 is that the naive statement “ $\sqrt{n}(\hat{\mathbf{A}}_n - \mathbf{A}_0)$ is asymptotically normal” is not merely false but meaningless: by non-identifiability the estimator $\hat{\mathbf{A}}_n$ produced by any algorithm is only determined up to a right multiplication by an element of $O(q)$, so that $\hat{\mathbf{A}}_n$ need not converge to $\mathbf{A}_0$ at all, and

only its orbit can. The correct object to look at is the intrinsic estimation error $\text{Exp}_{[\mathbf{A}_0, s_0]}^{-1}([\hat{\mathbf{A}}_n, \hat{s}_n]) \in T_{[\mathbf{A}_0, s_0]}M$ of [Sun et al.(2026)Sun, Lin, and Liu], which by Proposition 5 is the ordinary error *after optimal rotational alignment*. The goal of this section is to prove that this tangent vector is asymptotically linear in the sense of [Sun et al.(2026)Sun, Lin, and Liu], Lemma III.3, with influence operator $G_{[\mathbf{A}_0, s_0]}^{-1}s(\cdot; [\mathbf{A}_0, s_0])$, and consequently asymptotically normal with covariance operator $G_{[\mathbf{A}_0, s_0]}^{-1}$.

It will be convenient to name the horizontal lift. By Proposition 1 and the definition (1.4) of g , the restriction of $d\pi_{(\mathbf{A}, s)}$ to $H_{(\mathbf{A}, s)}$ is a linear isometry onto $T_{[\mathbf{A}, s]}M$; we denote its inverse by

$$\iota_{(\mathbf{A}, s)} := \left(d\pi_{(\mathbf{A}, s)}|_{H_{(\mathbf{A}, s)}} \right)^{-1} : T_{[\mathbf{A}, s]}M \longrightarrow H_{(\mathbf{A}, s)}, \quad \iota_{(\mathbf{A}, s)}(\mathbf{h}) =: (\mathbf{U}_h, b_h), \quad (1.34)$$

so that $\iota_{(\mathbf{A}, s)}$ is a *linear* isometry and $|\mathbf{h}|^2 = \text{tr}(\mathbf{U}_h \mathbf{U}_h^T) + b_h^2$. Finally, since $G_{[\mathbf{A}, s]}$ is a positive definite bilinear form on $T_{[\mathbf{A}, s]}M$ by Theorem 7, the map $\mathbf{h} \mapsto G_{[\mathbf{A}, s]}(\mathbf{h}, \cdot)$ is a linear isomorphism from $T_{[\mathbf{A}, s]}M$ onto its dual $T_{[\mathbf{A}, s]}^*M$; we write $G_{[\mathbf{A}, s]}^{-1}$ for its inverse, characterized by

$$G_{[\mathbf{A}, s]} \left(G_{[\mathbf{A}, s]}^{-1}(\varphi), \mathbf{h} \right) = \varphi(\mathbf{h}) \quad \text{for all } \varphi \in T_{[\mathbf{A}, s]}^*M \text{ and all } \mathbf{h} \in T_{[\mathbf{A}, s]}M. \quad (1.35)$$

Since the score $s(\mathbf{x}; [\mathbf{A}, s])$ of (1.20) is a linear functional on $T_{[\mathbf{A}, s]}M$, the composition $G_{[\mathbf{A}, s]}^{-1}s(\mathbf{x}; [\mathbf{A}, s])$ is a genuine tangent vector.

The maximum likelihood estimator

The log-likelihood of the model (1.6) is the function on M given by

$$\mathbb{L}_n([\mathbf{A}, s]) := \sum_{i=1}^n \log p(\mathbf{X}_i; [\mathbf{A}, s]) = -\frac{np}{2} \log(2\pi) - \frac{n}{2} \log \det \Sigma([\mathbf{A}, s]) - \frac{n}{2} \text{tr}((\Sigma([\mathbf{A}, s]))^{-1} \mathbf{S}_n); \quad (1.36)$$

it depends on the sample only through $\mathbf{S}_n$ and it is genuinely a function on the quotient M , since the right hand side depends on $(\mathbf{A}, s)$ only through Σ . This is precisely the reason why the model had to be quotiented in the first place: $\mathbb{L}_n$ is exactly constant along the fibers of π , so no maximizer of $\mathbb{L}_n$ on Θ can ever be unique, whereas on M it can. The following classical computation identifies the maximizer. We write $\lambda_1(\mathbf{S}) \geq \dots \geq \lambda_p(\mathbf{S})$ for the eigenvalues of $\mathbf{S} \in \text{Sym}(p)$, the space of real symmetric $p \times p$ matrices, repeated according to multiplicity, and we let

$$\mathcal{O} := \{\mathbf{S} \in \text{Sym}(p) : \lambda_q(\mathbf{S}) > \lambda_{q+1}(\mathbf{S})\}, \quad (1.37)$$

which is an open subset of $\text{Sym}(p)$ because the eigenvalues depend continuously on $\mathbf{S}$. For $\mathbf{S} \in \mathcal{O}$ we write $\mathbf{P}_q(\mathbf{S})$ for the orthogonal projector of $\mathbb{R}^p$ onto the sum of the eigenspaces of $\lambda_1(\mathbf{S}), \dots, \lambda_q(\mathbf{S})$; this is well defined precisely because of the spectral gap in (1.37).

Proposition 9 (Tipping–Bishop). Let $\mathbf{S} \in \mathcal{O}$ be positive definite and put $\lambda_j := \lambda_j(\mathbf{S})$ and $\mathbf{P}_q := \mathbf{P}_q(\mathbf{S})$. Then the function

$$[\mathbf{A}, s] \longmapsto -\frac{p}{2} \log(2\pi) - \frac{1}{2} \log \det \Sigma([\mathbf{A}, s]) - \frac{1}{2} \text{tr}((\Sigma([\mathbf{A}, s]))^{-1} \mathbf{S})$$

attains its maximum over M at one and only one point $[\hat{\mathbf{A}}, \hat{s}] \in M$, determined by

$$\hat{s} = \frac{1}{p-q} \sum_{j=q+1}^p \lambda_j, \quad \hat{\mathbf{A}} \hat{\mathbf{A}}^T = \mathbf{P}_q \mathbf{S} \mathbf{P}_q - \hat{s} \mathbf{P}_q. \quad (1.38)$$

Concretely, $\hat{\mathbf{A}} = \mathbf{W}_q(\Lambda_q - \hat{s} \mathbf{I}_q)^{\frac{1}{2}} \mathbf{R}$ for an arbitrary $\mathbf{R} \in O(q)$, where the columns of $\mathbf{W}_q \in \mathbb{R}^{p \times q}$ are orthonormal eigenvectors of $\mathbf{S}$ associated with $\lambda_1, \dots, \lambda_q$ and $\Lambda_q = \text{diag}(\lambda_1, \dots, \lambda_q)$.

Proof. This is Section 3.2 together with Appendix A of [Tipping and Bishop(1999)]: equations (7) and (8) there give the stationary points and equations (17)–(20) of their Appendix A.2 show that the global maximum is attained exactly when the q largest eigenvalues of $\mathbf{S}$ are retained. Since $\lambda_q > \lambda_{q+1}$ the set of retained eigenvalues, and hence the projector $\mathbf{P}_q$, is unambiguous, so the maximizer is a single point of M . The two descriptions agree because $\mathbf{P}_q \mathbf{S} \mathbf{P}_q = \mathbf{W}_q \mathbf{\Lambda}_q \mathbf{W}_q^T$ and $\mathbf{P}_q = \mathbf{W}_q \mathbf{W}_q^T$. Finally $[\hat{\mathbf{A}}, \hat{s}]$ is really a point of M : $\hat{s} > 0$ because $\mathbf{S}$ is positive definite, and $\text{rank}(\hat{\mathbf{A}}) = q$ because $\hat{s} \leq \lambda_{q+1} < \lambda_q$ forces every diagonal entry of $\mathbf{\Lambda}_q - \hat{s} \mathbf{I}_q$ to be strictly positive. $\square$

Definition 10. Whenever $\mathbf{S}_n$ is positive definite and lies in $\mathcal{O}$, we let $[\hat{\mathbf{A}}_n, \hat{s}_n] \in M$ denote the maximum likelihood estimator, that is the unique maximizer of (1.36) over M supplied by Proposition 9 applied to $\mathbf{S} = \mathbf{S}_n$. On the complementary event we set $[\hat{\mathbf{A}}_n, \hat{s}_n] := [\mathbf{A}_0, s_0]$; Lemma 12 below shows that this convention is invoked only on a null set as soon as $n \geq p$.

Genericity of the sample covariance matrix

Both the well-posedness of the estimation error and the consistency of the maximum likelihood estimator rest on the behaviour of the spectral projector $\mathbf{P}_q$, which we now record.

Lemma 11.

1. $\mathcal{O}$ is open and connected in $\text{Sym}(p)$.
2. $\mathbf{P}_q : \mathcal{O} \rightarrow \text{Sym}(p)$ is real analytic.
3. Let $\mathbf{U}_0 \in \mathbb{R}^{p \times q}$ have orthonormal columns and define $\phi : \mathcal{O} \rightarrow \mathbb{R}$ by $\phi(\mathbf{S}) := \det(\mathbf{U}_0^T \mathbf{P}_q(\mathbf{S}) \mathbf{U}_0)$. Then ϕ is real analytic and $\phi^{-1}(\{0\})$ is a Lebesgue null subset of $\text{Sym}(p) \cong \mathbb{R}^{\frac{p(p+1)}{2}}$.

Proof. 1. Openness was already observed. For connectedness, fix $\mathbf{S} \in \mathcal{O}$ and diagonalize $\mathbf{S} = \mathbf{K} \text{diag}(\boldsymbol{\lambda}) \mathbf{K}^T$ with $\mathbf{K} \in O(p)$ and $\boldsymbol{\lambda} = (\lambda_1(\mathbf{S}), \dots, \lambda_p(\mathbf{S}))$ non-increasing; replacing one column of $\mathbf{K}$ by its negative if necessary, which does not change $\mathbf{S}$, we may assume $\det(\mathbf{K}) = 1$. Let $\boldsymbol{\mu} := (2, \dots, 2, 1, \dots, 1)$ with the value 2 repeated q times, and consider first the path $t \mapsto \mathbf{K} \text{diag}((1-t)\boldsymbol{\lambda} + t\boldsymbol{\mu}) \mathbf{K}^T$ for $t \in [0, 1]$. The vector $(1-t)\boldsymbol{\lambda} + t\boldsymbol{\mu}$ is non-increasing, being a convex combination of non-increasing vectors, so it lists the ordered eigenvalues, and the gap at position q equals $(1-t)(\lambda_q(\mathbf{S}) - \lambda_{q+1}(\mathbf{S})) + t > 0$; the path therefore stays inside $\mathcal{O}$ and joins $\mathbf{S}$ to $\mathbf{K} \text{diag}(\boldsymbol{\mu}) \mathbf{K}^T$. Next, $SO(p)$ is path connected, so there is a path $u \mapsto \mathbf{K}(u)$ in $SO(p)$ from $\mathbf{K}$ to $\mathbf{I}_p$; the induced path $u \mapsto \mathbf{K}(u) \text{diag}(\boldsymbol{\mu}) \mathbf{K}(u)^T$ has constant spectrum $\boldsymbol{\mu}$, hence constant gap $1 > 0$ at position q , stays in $\mathcal{O}$, and ends at $\text{diag}(\boldsymbol{\mu})$. Every point of $\mathcal{O}$ is thus joined to the fixed point $\text{diag}(\boldsymbol{\mu})$ by a path inside $\mathcal{O}$.

2. Fix $\mathbf{S}_* \in \mathcal{O}$, choose $c \in \mathbb{R}$ with $\lambda_{q+1}(\mathbf{S}_*) < c < \lambda_q(\mathbf{S}_*)$ and let $\Gamma \subset \mathbb{C}$ be the positively oriented circle of centre $\frac{1}{2}(\lambda_1(\mathbf{S}_*) + c)$ and radius $\frac{1}{2}(\lambda_1(\mathbf{S}_*) - c) + 1$, so that Γ encircles $\lambda_1(\mathbf{S}_*), \dots, \lambda_q(\mathbf{S}_*)$ and no other eigenvalue of $\mathbf{S}_*$. Put $\delta := \min_{z \in \Gamma} \text{dist}(z, \text{spec}(\mathbf{S}_*)) > 0$. If $\mathbf{S} \in \text{Sym}(p)$ satisfies $\|\mathbf{S} - \mathbf{S}_*\|_2 < \delta$ then $z\mathbf{I}_p - \mathbf{S}$ is invertible for every $z \in \Gamma$, Weyl's inequality keeps exactly q eigenvalues of $\mathbf{S}$ inside Γ , and diagonalizing $\mathbf{S}$ shows, by the residue theorem applied to each eigenvalue separately, that

$$\mathbf{P}_q(\mathbf{S}) = \frac{1}{2\pi i} \oint_{\Gamma} (z\mathbf{I}_p - \mathbf{S})^{-1} dz. \quad (1.39)$$

Expanding the resolvent in a Neumann series,

$$(z\mathbf{I}_p - \mathbf{S})^{-1} = \sum_{k=0}^{\infty} [(z\mathbf{I}_p - \mathbf{S}_*)^{-1} (\mathbf{S} - \mathbf{S}_*)]^k (z\mathbf{I}_p - \mathbf{S}_*)^{-1},$$

a series which converges absolutely and uniformly in $z \in \Gamma$ whenever $\|\mathbf{S} - \mathbf{S}_*\|_2 < \delta$, since $\|(z\mathbf{I}_p - \mathbf{S}_*)^{-1}\|_2 \leq \delta^{-1}$ there. Each summand is a homogeneous polynomial of degree k in the entries of $\mathbf{S} - \mathbf{S}_*$ whose coefficients

are continuous in z , so integrating term by term over the compact contour Γ exhibits P_q as a convergent power series in the entries of $S - S_*$ on that ball.

3. Real analyticity of ϕ is immediate from 2, the determinant being a polynomial. Moreover ϕ does not vanish identically: taking $S_* := I_p + U_0 U_0^T$, whose eigenvalues are 2 with multiplicity q , with eigenspace the column space of U_0 , and 1 with multiplicity $p - q$, we get $S_* \in \mathcal{O}$, $P_q(S_*) = U_0 U_0^T$ and $\phi(S_*) = \det(I_q) = 1$. By 1 the domain $\mathcal{O}$ is open and connected, so the zero set of the non-trivial real analytic function ϕ is Lebesgue null in $\text{Sym}(p)$ by [Mityagin(2020)]. Finally $\text{Sym}(p) \setminus \mathcal{O}$ is itself Lebesgue null: it is contained in the zero set of the discriminant of the characteristic polynomial of S , which is a polynomial in the entries of S and does not vanish at $\text{diag}(1, 2, \dots, p)$. $\square$

Lemma 12. Suppose $n \geq p$. Then, almost surely, S_n is positive definite, $S_n \in \mathcal{O}$, and the $q \times q$ matrix $A_0^T \hat{A}_n$ is invertible. In particular the orthogonal matrix

$$Q_n := Q(A_0, \hat{A}_n) = \left(\hat{A}_n^T A_0 A_0^T \hat{A}_n \right)^{-\frac{1}{2}} \hat{A}_n^T A_0 \in O(q) \quad (1.40)$$

of Proposition 5 is almost surely well defined.

Proof. We first check that the law of S_n is absolutely continuous with respect to the Lebesgue measure of $\text{Sym}(p)$. Let $Y_1 \in \mathbb{R}^{p \times p}$ be the matrix whose rows are $X_1^T, \dots, X_p^T$ and let Y_2 collect the remaining $n - p$ observations, so that $S_n = \frac{1}{n} (Y_1^T Y_1 + Y_2^T Y_2)$ and Y_1 is independent of Y_2 with a strictly positive Lebesgue density on $\mathbb{R}^{p \times p}$. Condition on Y_2 and consider $T(Y) := \frac{1}{n} (Y^T Y + Y_2^T Y_2)$ on the open set of invertible $Y \in \mathbb{R}^{p \times p}$, whose complement is Lebesgue null. Its differential at Y is $H \mapsto \frac{1}{n} (H^T Y + Y^T H)$, which is onto $\text{Sym}(p)$: given $B \in \text{Sym}(p)$, the choice $H = \frac{n}{2} Y^{-T} B$ returns B . Thus T is a smooth submersion, and by the rank theorem ([Lee(2013)], Theorem 4.12) every point has a neighbourhood on which T is, up to diffeomorphisms, a linear projection $\mathbb{R}^{p^2} \rightarrow \mathbb{R}^{\frac{p(p+1)}{2}}$. Since preimages of Lebesgue null sets under such projections are Lebesgue null by Fubini's theorem, and diffeomorphisms preserve Lebesgue null sets, $T^{-1}(N)$ is Lebesgue null for every Lebesgue null $N \subseteq \text{Sym}(p)$, after covering the domain by countably many such neighbourhoods. Hence the conditional law of S_n given Y_2 charges no Lebesgue null set, and the same holds for its mixture over Y_2 .

By Lemma 11, the set of $S \in \text{Sym}(p)$ that either fail to be positive definite with p distinct eigenvalues, or lie outside $\mathcal{O}$, or satisfy $\phi(S) = 0$, is Lebesgue null; here we take $U_0 \in \mathbb{R}^{p \times q}$ to be any matrix with orthonormal columns spanning the column space of A_0 , which has dimension q because $\text{rank}(A_0) = q$. It follows that almost surely S_n is positive definite, $S_n \in \mathcal{O}$ and $\phi(S_n) \neq 0$.

It remains to see that $\phi(S_n) \neq 0$ is exactly the invertibility of $A_0^T \hat{A}_n$. Write $A_0 = U_0 T$ with $T \in \mathbb{R}^{q \times q}$ invertible, which is possible since the columns of U_0 form a basis of the column space of A_0 , and use the representative $\hat{A}_n = W_q (\Lambda_q - \hat{s}_n I_q)^{\frac{1}{2}}$ of Proposition 9; note that invertibility of $A_0^T \hat{A}_n$ is unchanged if $\hat{A}_n$ is replaced by $\hat{A}_n R$ with $R \in O(q)$, so it is a property of the class $[\hat{A}_n, \hat{s}_n]$. Since $\Lambda_q - \hat{s}_n I_q$ is invertible, we get

$$A_0^T \hat{A}_n \text{ is invertible} \iff U_0^T W_q \text{ is invertible} \iff \det(U_0^T W_q W_q^T U_0) \neq 0 \iff \phi(S_n) \neq 0,$$

where we used $\det(U_0^T W_q)^2 = \det(U_0^T W_q W_q^T U_0)$ and $W_q W_q^T = P_q(S_n)$. $\square$

Consistency and the aligned estimation error

Proposition 13. Almost surely, $[\hat{A}_n, \hat{s}_n] \rightarrow [A_0, s_0]$ in M .

Proof. Write P_0 for the orthogonal projector onto the column space of A_0 , so that $P_0 A_0 = A_0$. The eigenvalues of $\Sigma_0 = A_0 A_0^T + s_0 I_p$ are $\sigma_j(A_0)^2 + s_0$ for $j = 1, \dots, q$ and s_0 with multiplicity $p - q$, where σ_j denotes the singular values; since $\text{rank}(A_0) = q$ we have $\sigma_q(A_0) > 0$ and hence $\lambda_q(\Sigma_0) > s_0 = \lambda_{q+1}(\Sigma_0)$. Thus $\Sigma_0 \in \mathcal{O}$ and $P_q(\Sigma_0) = P_0$.

By the strong law of large numbers applied to each of the finitely many entries, $S_n \rightarrow \Sigma_0$ almost surely. As $\mathcal{O}$ is open and P_q is continuous on $\mathcal{O}$ by Lemma 11, almost surely $S_n \in \mathcal{O}$ for all large n and $P_q(S_n) \rightarrow P_0$. Since S_n is eventually positive definite as well, Proposition 9 applies and (1.38) gives, almost surely,

$$\hat{s}_n = \frac{1}{p-q} \operatorname{tr}((I_p - P_q(S_n))S_n) \longrightarrow \frac{1}{p-q} \operatorname{tr}((I_p - P_0)\Sigma_0) = \frac{s_0}{p-q} \operatorname{tr}(I_p - P_0) = s_0$$

because $(I_p - P_0)\Sigma_0 = s_0(I_p - P_0)$, and

$$\hat{A}_n \hat{A}_n^T = P_q(S_n)S_n P_q(S_n) - \hat{s}_n P_q(S_n) \longrightarrow P_0 \Sigma_0 P_0 - s_0 P_0 = A_0 A_0^T,$$

using $P_0 \Sigma_0 P_0 = A_0 A_0^T + s_0 P_0$.

It remains to convert this into convergence in M . We claim that if $B_n \in \mathbb{R}^{p \times q}$ satisfy $B_n B_n^T \rightarrow A_0 A_0^T$, then $\min_{Q \in O(q)} \|B_n Q - A_0\|_F \rightarrow 0$. Indeed $\|B_n\|_F^2 = \operatorname{tr}(B_n B_n^T)$ is convergent, hence bounded. If the claim failed there would be $\epsilon > 0$ and a subsequence along which $\min_Q \|B_n Q - A_0\|_F \geq \epsilon$; by boundedness we may extract a further subsequence with $B_n \rightarrow B$, and then $B B^T = A_0 A_0^T$. This matrix has rank q , so $\operatorname{rank}(B) = q$ and $(B, s_0) \sim (A_0, s_0)$; by the description of the equivalence classes recalled in Section 1.1 there is $Q_0 \in O(q)$ with $B = A_0 Q_0$. Consequently $\min_Q \|B_n Q - A_0\|_F \leq \|B_n Q_0^T - B Q_0^T\|_F = \|B_n - B\|_F \rightarrow 0$ along that subsequence, a contradiction.

Applying the claim to $B_n = \hat{A}_n$ we obtain, almost surely, matrices $Q'_n \in O(q)$ with $\hat{A}_n Q'_n \rightarrow A_0$; combined with $\hat{s}_n \rightarrow s_0$ this says $(\hat{A}_n Q'_n, \hat{s}_n) \rightarrow (A_0, s_0)$ in Θ , and the continuity of π gives

$$[\hat{A}_n, \hat{s}_n] = \pi\left((\hat{A}_n Q'_n, \hat{s}_n)\right) \longrightarrow \pi((A_0, s_0)) = [A_0, s_0]$$

in M , as required. $\square$

Remark 14. Proposition 13 recovers, for the independent and identically distributed Gaussian sampling scheme, the strong consistency result of [Datta and Chakrabarty(2023)] and [Datta(2026)], Corollary 1, which is proved there by a Wald-type argument valid under considerably weaker assumptions. The proof above is included because it is short once Lemma 11 is available, and because it makes the section self-contained.

We can now fix once and for all the normal chart in which the whole analysis will be carried out.

Lemma 15. There exists $\rho \in (0, \min(\sigma_{\min}(A_0), s_0))$ such that, writing $B_\rho := \{u \in T_{[A_0, s_0]} M : |u| < \rho\} \subset \mathcal{E}_{[A_0, s_0]}$, the map $\operatorname{Exp}_{[A_0, s_0]}$ restricted to B_ρ is a diffeomorphism onto an open subset $\mathcal{U}_0 := \operatorname{Exp}_{[A_0, s_0]}(B_\rho) \subseteq M$, and such that $\mathcal{U}_0 \subseteq U$, where U is the neighbourhood supplied by Proposition 5 at $[A_0, s_0]$.

Proof. By Proposition 2 the exponential map is defined on $\mathcal{E}_{[A_0, s_0]}$, and it is a local diffeomorphism at $\mathbf{0}$, so there is an open neighbourhood $\mathcal{N}$ of $\mathbf{0}$ inside $\mathcal{E}_{[A_0, s_0]}$ that is mapped diffeomorphically onto an open subset of M . Since $\operatorname{Exp}_{[A_0, s_0]}$ is continuous and $\operatorname{Exp}_{[A_0, s_0]}(\mathbf{0}) = [A_0, s_0] \in U$, we may pick $\rho > 0$ so small that $B_\rho \subseteq \mathcal{N}$ and $\operatorname{Exp}_{[A_0, s_0]}(B_\rho) \subseteq U$. A restriction of a diffeomorphism to an open subset is again a diffeomorphism onto its (open) image. $\square$

We fix such a ρ for the rest of the section, and define the *aligned estimation error*

$$e_n := (\hat{A}_n Q_n - A_0, \hat{s}_n - s_0) \in \mathbb{R}^{p \times q} \times \mathbb{R}, \quad \hat{u}_n := \begin{cases} \operatorname{Exp}_{[A_0, s_0]}^{-1}([\hat{A}_n, \hat{s}_n]) & \text{on } \mathcal{A}_n, \\ \mathbf{0} & \text{otherwise,} \end{cases} \quad (1.41)$$

where Q_n is as in (1.40) and $\mathcal{A}_n := \{[\hat{A}_n, \hat{s}_n] \in \mathcal{U}_0\}$.

Proposition 16. For $n \geq p$ we have $\mathbb{P}(\mathcal{A}_n) \rightarrow 1$; on $\mathcal{A}_n$ the vector $\hat{u}_n$ belongs to B_ρ and its horizontal lift at (A_0, s_0) is exactly e_n , that is $\iota_{(A_0, s_0)}(\hat{u}_n) = e_n$; and $\hat{u}_n \rightarrow \mathbf{0}$ almost surely.

Proof. By Lemma 12 the matrix $\mathbf{Q}_n$ is almost surely well defined for $n \geq p$, so $\mathbf{e}_n$ is. By Proposition 13 and the fact that $\mathcal{U}_0$ is an open neighbourhood of $[\mathbf{A}_0, s_0]$, almost surely $[\hat{\mathbf{A}}_n, \hat{s}_n] \in \mathcal{U}_0$ for all large n ; in particular $\mathbb{P}(\mathcal{A}_n) \rightarrow 1$. On $\mathcal{A}_n$ we have $\hat{\mathbf{u}}_n \in B_\rho$ by Lemma 15, and since $\mathcal{U}_0 \subseteq U$ the formula of Proposition 5 applies and reads

$$\hat{\mathbf{u}}_n = d\pi_{(\mathbf{A}_0, s_0)} \left(\left(\hat{\mathbf{A}}_n \mathbf{Q}_n - \mathbf{A}_0, \hat{s}_n - s_0 \right) \right) = d\pi_{(\mathbf{A}_0, s_0)}(\mathbf{e}_n).$$

As observed in the proof of Proposition 5, $\mathbf{e}_n$ is a horizontal vector at $(\mathbf{A}_0, s_0)$, so $\iota_{(\mathbf{A}_0, s_0)}(\hat{\mathbf{u}}_n) = \mathbf{e}_n$ by (1.34). Finally let $\epsilon \in (0, \rho)$. Then $\text{Exp}_{[\mathbf{A}_0, s_0]}(B_\epsilon)$ is an open neighbourhood of $[\mathbf{A}_0, s_0]$, so almost surely $[\hat{\mathbf{A}}_n, \hat{s}_n]$ lies in it for large n , and injectivity of $\text{Exp}_{[\mathbf{A}_0, s_0]}$ on B_ρ then forces $|\hat{\mathbf{u}}_n| < \epsilon$. $\square$

Remark 17. Assumption III.2 of [Sun et al.(2026)Sun, Lin, and Liu] postulates that the estimator almost surely avoids the cut locus of the true parameter, so that the logarithmic map may be applied to it, and Proposition 16 is what plays that role here. We cannot verify that assumption literally, because M is incomplete and the discussion following Proposition 5 shows that $\text{Exp}_{[\mathbf{A}_0, s_0]}^{-1}$ genuinely fails to be globally defined. What Proposition 16 provides instead is the only thing the asymptotic theory ever uses: an event of probability tending to one on which the intrinsic error is well defined, together with the explicit Procrustes description (1.41) of it. Since modifying a sequence of random vectors on a sequence of events of vanishing probability affects neither its limit in probability nor its limit in distribution, all statements below are insensitive to the convention adopted off $\mathcal{A}_n$.

Asymptotic linearity

We can now state the definition, which is Definition 3.10 of the companion draft and, in the language of [Sun et al.(2026)Sun, Lin, and Liu], the hypothesis of their Lemma III.3.

Definition 18. Let $[\hat{\mathbf{A}}_n, \hat{s}_n]$ be a sequence of estimators for which the intrinsic estimation error $\hat{\mathbf{u}}_n$ of (1.41) is defined on events of probability tending to one. The sequence is said to be *asymptotically linear at $[\mathbf{A}_0, s_0]$ with influence operator IF* if $\text{IF} : \mathbb{R}^p \rightarrow T_{[\mathbf{A}_0, s_0]}M$ is measurable, centred and square integrable, that is

$$\mathbb{E}[\text{IF}(\mathbf{X}_1)] = \mathbf{0} \quad \text{and} \quad \mathbb{E}[|\text{IF}(\mathbf{X}_1)|^2] < \infty, \quad (1.42)$$

and if

$$\sqrt{n} \hat{\mathbf{u}}_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n \text{IF}(\mathbf{X}_i) + o_{\mathbb{P}}(1), \quad (1.43)$$

the remainder being $o_{\mathbb{P}}(1)$ with respect to the norm $|\cdot|$ of $T_{[\mathbf{A}_0, s_0]}M$.

Remark 19. Both sides of (1.43) live in the single fixed vector space $T_{[\mathbf{A}_0, s_0]}M$, so the expression is meaningful even though M is curved; this is exactly the device by which [Sun et al.(2026)Sun, Lin, and Liu] transport the classical theory to manifolds. Note also that IF is required to take values in the *tangent* space, whereas the score (1.20) is a linear functional, that is an element of the *cotangent* space; the inverse Fisher operator (1.35) is precisely what converts one into the other.

The following theorem is the main result of this section. It is the analogue of Theorem 5.3 of the companion draft, and it instantiates Corollary V.1 of [Sun et al.(2026)Sun, Lin, and Liu] for the probabilistic principal component analysis model.

Theorem 20. Let $1 \leq q < p$ and let $[\mathbf{A}_0, s_0] \in M$. Then the maximum likelihood estimator $[\hat{\mathbf{A}}_n, \hat{s}_n]$ of Definition 10 is asymptotically linear at $[\mathbf{A}_0, s_0]$ in the sense of Definition 18, with influence operator

$$\text{IF}(\mathbf{x}) := G_{[\mathbf{A}_0, s_0]}^{-1}(s(\mathbf{x}; [\mathbf{A}_0, s_0])) \in T_{[\mathbf{A}_0, s_0]}M. \quad (1.44)$$

That is,

$$\sqrt{n} \text{Exp}_{[\mathbf{A}_0, s_0]}^{-1}([\hat{\mathbf{A}}_n, \hat{s}_n]) = \frac{1}{\sqrt{n}} \sum_{i=1}^n G_{[\mathbf{A}_0, s_0]}^{-1}(s(\mathbf{X}_i; [\mathbf{A}_0, s_0])) + o_{\mathbb{P}}(1), \quad (1.45)$$

and consequently

$$\sqrt{n} \text{Exp}_{[\mathbf{A}_0, s_0]}^{-1} \left([\hat{\mathbf{A}}_n, \hat{s}_n] \right) \rightsquigarrow N \left(\mathbf{0}, G_{[\mathbf{A}_0, s_0]}^{-1} \right) \quad \text{on } T_{[\mathbf{A}_0, s_0]} M, \quad (1.46)$$

the Gaussian measure on $T_{[\mathbf{A}_0, s_0]} M$ whose covariance operator is the inverse Fisher information operator. Equivalently, in terms of the Procrustes aligned error (1.41),

$$\sqrt{n} \left(\hat{\mathbf{A}}_n \mathbf{Q}_n - \mathbf{A}_0, \hat{s}_n - s_0 \right) \rightsquigarrow \iota_{(\mathbf{A}_0, s_0)} \left(N \left(\mathbf{0}, G_{[\mathbf{A}_0, s_0]}^{-1} \right) \right) \quad (1.47)$$

as random elements of $\mathbb{R}^{p \times q} \times \mathbb{R}$; the limit is a centred Gaussian law carried by the d -dimensional subspace $H_{(\mathbf{A}_0, s_0)}$.

Proof. Write $G := G_{[\mathbf{A}_0, s_0]}$, $\iota := \iota_{(\mathbf{A}_0, s_0)}$ and $s(\mathbf{x}) := s(\mathbf{x}; [\mathbf{A}_0, s_0])$ throughout. The strategy is the one of Step 2 of the companion draft: in the normal chart supplied by Lemma 15 the family $\mathcal{P}$ becomes an ordinary parametric family indexed by an open subset of a Euclidean space, and $\hat{\mathbf{u}}_n$ becomes an ordinary M -estimator, so that Theorem 5.23 of [van der Vaart(1998)] applies once its four hypotheses have been verified. The essential structural point, which makes every verification a finite computation, is that by Proposition 2 the exponential map is linear in the horizontal lift, so no approximation of Exp is ever needed.

Step 1: the model in normal coordinates. Fix a $g_{[\mathbf{A}_0, s_0]}$ -orthonormal basis $\mathbf{e}_1, \dots, \mathbf{e}_d$ of $T_{[\mathbf{A}_0, s_0]} M$ and identify $T_{[\mathbf{A}_0, s_0]} M$ with $\mathbb{R}^d$ through it; under this identification B_ρ is the open Euclidean ball of radius ρ and $|\cdot|$ is the Euclidean norm. For $\mathbf{u} \in B_\rho$ put $(\mathbf{U}_u, b_u) := \iota(\mathbf{u})$ and

$$\Sigma_u := \Sigma \left(\text{Exp}_{[\mathbf{A}_0, s_0]}(\mathbf{u}) \right) = (\mathbf{A}_0 + \mathbf{U}_u)(\mathbf{A}_0 + \mathbf{U}_u)^T + (s_0 + b_u)\mathbf{I}_p, \quad (1.48)$$

using Proposition 2 and (1.5). Because ι is linear, expanding (1.48) gives the *exact* identity

$$\Sigma_u = \Sigma_0 + (\mathbf{U}_u \mathbf{A}_0^T + \mathbf{A}_0 \mathbf{U}_u^T + b_u \mathbf{I}_p) + \mathbf{U}_u \mathbf{U}_u^T = \Sigma_0 + D_u \Sigma([\mathbf{A}_0, s_0]) + \mathbf{U}_u \mathbf{U}_u^T, \quad (1.49)$$

by (1.21): as a function of $\mathbf{u}$, the covariance matrix is a quadratic polynomial with no remainder whatsoever. Set, for $\mathbf{u} \in B_\rho$ and $\mathbf{x} \in \mathbb{R}^p$,

$$m(\mathbf{u}; \mathbf{x}) := \log p \left(\mathbf{x}; \text{Exp}_{[\mathbf{A}_0, s_0]}(\mathbf{u}) \right) = -\frac{p}{2} \log(2\pi) - \frac{1}{2} \log \det \Sigma_u - \frac{1}{2} \mathbf{x}^T \Sigma_u^{-1} \mathbf{x}, \quad (1.50)$$

which in the notation (1.18) is $l_{[\mathbf{A}_0, s_0]}(\mathbf{u})(\mathbf{x})$, and write $\mathbb{M}_n(\mathbf{u}) := \frac{1}{n} \sum_{i=1}^n m(\mathbf{u}; \mathbf{X}_i)$ and $M(\mathbf{u}) := \mathbb{E}[m(\mathbf{u}; \mathbf{X}_1)]$. By (1.36) and Lemma 15, on the event $\mathcal{A}_n$ the vector $\hat{\mathbf{u}}_n$ maximizes $\mathbb{M}_n$ over B_ρ , since $[\hat{\mathbf{A}}_n, \hat{s}_n]$ maximizes $\mathbb{L}_n$ over all of M and lies in the chart.

Step 2: uniform spectral bounds. Let $\mathbf{u} \in B_\rho$. Since ι is an isometry, $\|\mathbf{U}_u\|_F \leq \rho$ and $|b_u| \leq \rho$; as $\rho < s_0$, formula (1.48) gives

$$c_1 \mathbf{I}_p \preceq \Sigma_u \preceq c_2 \mathbf{I}_p, \quad c_1 := s_0 - \rho > 0, \quad c_2 := (\sigma_{\max}(\mathbf{A}_0) + \rho)^2 + s_0 + \rho, \quad (1.51)$$

the lower bound because $(\mathbf{A}_0 + \mathbf{U}_u)(\mathbf{A}_0 + \mathbf{U}_u)^T \succeq \mathbf{0}$, and the upper bound because $\sigma_{\max}(\mathbf{A}_0 + \mathbf{U}_u) \leq \sigma_{\max}(\mathbf{A}_0) + \|\mathbf{U}_u\|_F$. In particular $\|\Sigma_u^{-1}\|_2 \leq c_1^{-1}$ and $\|\Sigma_u\|_F \leq \sqrt{p} c_2$ uniformly on B_ρ .

Step 3: smoothness and the derivative of the criterion. On the open cone of positive definite matrices the maps $\Sigma \mapsto \Sigma^{-1}$ and $\Sigma \mapsto \log \det \Sigma$ are real analytic, and $\mathbf{u} \mapsto \Sigma_u$ is a polynomial by (1.49); hence for every fixed $\mathbf{x}$ the function $\mathbf{u} \mapsto m(\mathbf{u}; \mathbf{x})$ is C^∞ on B_ρ . Its differential is computed exactly as in (1.24): for $\mathbf{h} \in T_{[\mathbf{A}_0, s_0]} M$,

$$\nabla m(\mathbf{u}; \mathbf{x})[\mathbf{h}] = \frac{1}{2} \text{tr} \left\{ \Sigma_u^{-1} [\mathbf{x} \mathbf{x}^T - \Sigma_u] \Sigma_u^{-1} \left(D_h \Sigma \left(\text{Exp}_{[\mathbf{A}_0, s_0]}(\mathbf{u}) \right) \right) \right\}, \quad (1.52)$$

where, by (1.23),

$$D_h \Sigma \left(\text{Exp}_{[\mathbf{A}_0, s_0]}(\mathbf{u}) \right) = \mathbf{U}_h (\mathbf{A}_0 + \mathbf{U}_u)^T + (\mathbf{A}_0 + \mathbf{U}_u) \mathbf{U}_h^T + b_h \mathbf{I}_p, \quad (1.53)$$

which is linear in $\mathbf{h}$ and affine in $\mathbf{u}$. Taking $\mathbf{u} = \mathbf{0}$ in (1.52) and comparing with (1.20) shows

$$\nabla m(\mathbf{0}; \mathbf{x})[\mathbf{h}] = s(\mathbf{x})(\mathbf{h}), \quad (1.54)$$

so that the gradient of the criterion at the truth is the score.

Step 4: a square integrable Lipschitz envelope. Let $\mathbf{u} \in B_\rho$ and let $\mathbf{h}$ be a unit vector. From (1.53) and $\|\mathbf{U}_\mathbf{h} \mathbf{C}^T\|_F \leq \|\mathbf{U}_\mathbf{h}\|_F \|\mathbf{C}\|_2$,

$$\left\| D_\mathbf{h} \Sigma \left(\text{Exp}_{[\mathbf{A}_0, s_0]}(\mathbf{u}) \right) \right\|_F \leq 2(\sigma_{\max}(\mathbf{A}_0) + \rho) + \sqrt{p} =: c_3.$$

Combining this with $|\text{tr}(\mathbf{Z}\mathbf{Y})| \leq \|\mathbf{Z}\|_F \|\mathbf{Y}\|_F$ and the bounds of Step 2, (1.52) gives

$$\sup_{|\mathbf{h}|=1} |\nabla m(\mathbf{u}; \mathbf{x})[\mathbf{h}]| \leq \frac{c_3}{2c_1^2} (\|\mathbf{x}\|^2 + \sqrt{p}c_2) =: \dot{m}(\mathbf{x}) \quad \text{for all } \mathbf{u} \in B_\rho. \quad (1.55)$$

Since $\mathbf{X}_1$ is Gaussian, $\mathbb{E}[\|\mathbf{X}_1\|^4] < \infty$ and therefore $\mathbb{E}[\dot{m}(\mathbf{X}_1)^2] < \infty$. As B_ρ is convex and $m(\cdot; \mathbf{x})$ is C^1 there, the mean value theorem yields

$$|m(\mathbf{u}_1; \mathbf{x}) - m(\mathbf{u}_2; \mathbf{x})| \leq \dot{m}(\mathbf{x}) \|\mathbf{u}_1 - \mathbf{u}_2\| \quad \text{for all } \mathbf{u}_1, \mathbf{u}_2 \in B_\rho. \quad (1.56)$$

Step 5: the population criterion. Since $\mathbb{E}[\mathbf{X}_1 \mathbf{X}_1^T] = \Sigma_0$, (1.50) gives the closed form

$$M(\mathbf{u}) = -\frac{p}{2} \log(2\pi) - \frac{1}{2} \log \det \Sigma_\mathbf{u} - \frac{1}{2} \text{tr}(\Sigma_\mathbf{u}^{-1} \Sigma_0), \quad (1.57)$$

which, by the argument of Step 3, is C^∞ on B_ρ ; in particular no differentiation under the integral sign is needed anywhere below. Subtracting $M(\mathbf{0})$,

$$M(\mathbf{0}) - M(\mathbf{u}) = \frac{1}{2} \left[\text{tr}(\Sigma_\mathbf{u}^{-1} \Sigma_0) - p + \log \frac{\det \Sigma_\mathbf{u}}{\det \Sigma_0} \right] = \frac{1}{2} \sum_{j=1}^p (\mu_j - 1 - \log \mu_j), \quad (1.58)$$

where $\mu_1, \dots, \mu_p > 0$ are the eigenvalues of the positive definite matrix $\Sigma_\mathbf{u}^{-\frac{1}{2}} \Sigma_0 \Sigma_\mathbf{u}^{-\frac{1}{2}}$, which is similar to $\Sigma_\mathbf{u}^{-1} \Sigma_0$. The elementary inequality $\mu - 1 - \log \mu \geq 0$, with equality only at $\mu = 1$, shows that $M(\mathbf{0}) - M(\mathbf{u}) \geq 0$ with equality if and only if all $\mu_j = 1$, that is if and only if $\Sigma_\mathbf{u} = \Sigma_0$. By the injectivity of Σ established in Proposition 1 this happens exactly when $\text{Exp}_{[\mathbf{A}_0, s_0]}(\mathbf{u}) = [\mathbf{A}_0, s_0]$, and by Lemma 15 exactly when $\mathbf{u} = \mathbf{0}$. Hence $\mathbf{0}$ is the unique maximizer of M over B_ρ .

We now compute the second order Taylor expansion of M at $\mathbf{0}$. Fix $\mathbf{h} \in T_{[\mathbf{A}_0, s_0]} M$, write $(\mathbf{U}, b) := \iota(\mathbf{h})$, and let $\Sigma(t) := \Sigma_{t\mathbf{h}}$. By (1.49),

$$\Sigma(t) = \Sigma_0 + t\Delta + t^2 \mathbf{U} \mathbf{U}^T, \quad \Delta := D_\mathbf{h} \Sigma([\mathbf{A}_0, s_0]) = \mathbf{U} \mathbf{A}_0^T + \mathbf{A}_0 \mathbf{U}^T + b \mathbf{I}_p, \quad (1.59)$$

so $\dot{\Sigma}(0) = \Delta$ and $\ddot{\Sigma}(0) = 2\mathbf{U} \mathbf{U}^T$ exactly. Using $\frac{d}{dt} \log \det \Sigma = \text{tr}(\Sigma^{-1} \dot{\Sigma})$ and $\frac{d}{dt} \Sigma^{-1} = -\Sigma^{-1} \dot{\Sigma} \Sigma^{-1}$, the two terms of (1.57) contribute

$$\frac{d}{dt} \Big|_{t=0} \left[-\frac{1}{2} \log \det \Sigma(t) \right] = -\frac{1}{2} \text{tr}(\Sigma_0^{-1} \Delta), \quad \frac{d}{dt} \Big|_{t=0} \left[-\frac{1}{2} \text{tr}(\Sigma(t)^{-1} \Sigma_0) \right] = \frac{1}{2} \text{tr}(\Sigma_0^{-1} \Delta),$$

whose sum vanishes, so $\nabla M(\mathbf{0}) = 0$, as it must since $\mathbf{0}$ is an interior maximizer. Differentiating once more,

$$\frac{d^2}{dt^2} \Big|_{t=0} \left[-\frac{1}{2} \log \det \Sigma(t) \right] = -\frac{1}{2} \text{tr}(\Sigma_0^{-1} \ddot{\Sigma}(0)) + \frac{1}{2} \text{tr}((\Sigma_0^{-1} \Delta)^2)$$

and, since $\frac{d}{dt} \left(\Sigma^{-1} \dot{\Sigma} \Sigma^{-1} \right) = \Sigma^{-1} \ddot{\Sigma} \Sigma^{-1} - 2 \Sigma^{-1} \dot{\Sigma} \Sigma^{-1} \dot{\Sigma} \Sigma^{-1}$,

$$\frac{d^2}{dt^2} \Big|_{t=0} \left[-\frac{1}{2} \text{tr} \left(\Sigma(t)^{-1} \Sigma_0 \right) \right] = \frac{1}{2} \text{tr} \left(\Sigma_0^{-1} \ddot{\Sigma}(0) \right) - \text{tr} \left(\left(\Sigma_0^{-1} \Delta \right)^2 \right).$$

The terms in $\ddot{\Sigma}(0)$ cancel identically and we are left with

$$\frac{d^2}{dt^2} \Big|_{t=0} M(t\mathbf{h}) = -\frac{1}{2} \text{tr} \left(\left(\Sigma_0^{-1} D_{\mathbf{h}} \Sigma([\mathbf{A}_0, s_0]) \right)^2 \right) = -G(\mathbf{h}, \mathbf{h}) \quad (1.60)$$

by (1.22). As M is twice continuously differentiable on B_ρ , its Hessian at $\mathbf{0}$ is the symmetric bilinear form determined by the quadratic form (1.60), namely $-G$, and Taylor's theorem gives

$$M(\mathbf{u}) = M(\mathbf{0}) - \frac{1}{2} G(\mathbf{u}, \mathbf{u}) + o(|\mathbf{u}|^2) \quad \text{as } \mathbf{u} \rightarrow \mathbf{0}. \quad (1.61)$$

By Theorem 7 the form G is positive definite, so the second derivative matrix of M at $\mathbf{0}$ is symmetric and nonsingular. Incidentally, (1.60) is the manifold version of the classical identity between the Fisher information and the expected negative Hessian of the log-likelihood, recorded as Remark III.4 of [Sun et al.(2026)Sun, Lin, and Liu]; here it comes out of (1.49) by a finite computation precisely because the exponential map is linear in the horizontal lift.

Step 6: application of the M -estimation theorem. We check the four hypotheses of Theorem 5.23 of [van der Vaart(1998)] for the criterion functions $\{m(\mathbf{u}; \cdot)\}_{\mathbf{u} \in B_\rho}$ indexed by the open subset B_ρ of $\mathbb{R}^d$, at the point $\mathbf{u} = \mathbf{0}$.

1. For each $\mathbf{u}$ the map $\mathbf{x} \mapsto m(\mathbf{u}; \mathbf{x})$ is measurable, being continuous, and for each $\mathbf{x}$ the map $\mathbf{u} \mapsto m(\mathbf{u}; \mathbf{x})$ is differentiable at $\mathbf{0}$ with derivative $s(\mathbf{x}) \in L^2$, by Step 3 and (1.54); square integrability of the score is (1.28).
2. The Lipschitz condition with square integrable envelope is (1.56) and (1.55).
3. The map $\mathbf{u} \mapsto M(\mathbf{u})$ admits the second order Taylor expansion (1.61) at the maximizer $\mathbf{0}$, with nonsingular symmetric second derivative matrix $-G$, by Step 5.
4. By Proposition 16, $\hat{\mathbf{u}}_n \rightarrow \mathbf{0}$ in probability, and $\mathbb{M}_n(\hat{\mathbf{u}}_n) \geq \mathbb{M}_n(\mathbf{0})$ holds exactly: on $\mathcal{A}_n$ because $\hat{\mathbf{u}}_n$ maximizes $\mathbb{M}_n$ over $B_\rho \ni \mathbf{0}$ by Step 1, and off $\mathcal{A}_n$ because there $\hat{\mathbf{u}}_n = \mathbf{0}$. In particular $\mathbb{M}_n(\hat{\mathbf{u}}_n) \geq \mathbb{M}_n(\mathbf{0}) - o_{\mathbb{P}}(n^{-1})$.

The cited theorem therefore yields

$$\sqrt{n} \hat{\mathbf{u}}_n = \mathbf{G}^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n \mathbf{s}(\mathbf{X}_i) + o_{\mathbb{P}}(1), \quad (1.62)$$

where, in the orthonormal coordinates of Step 1, $\mathbf{G}$ is the $d \times d$ matrix with entries $\mathbf{G}_{jk} = G(\mathbf{e}_j, \mathbf{e}_k)$ and $\mathbf{s}(\mathbf{x}) \in \mathbb{R}^d$ is the vector with entries $\mathbf{s}(\mathbf{x})_j = s(\mathbf{x})(\mathbf{e}_j)$. Now if $\text{IF}(\mathbf{x}) = \sum_j c_j(\mathbf{x}) \mathbf{e}_j$ is defined by (1.44), then (1.35) reads $\sum_k \mathbf{G}_{jk} c_k(\mathbf{x}) = s(\mathbf{x})(\mathbf{e}_j)$ for every j , that is $\mathbf{c}(\mathbf{x}) = \mathbf{G}^{-1} \mathbf{s}(\mathbf{x})$; so (1.62) is precisely (1.45) and (1.43).

The requirements (1.42) hold: the score has mean zero and covariance G by Theorem 7, that is $\mathbb{E}[\mathbf{s}(\mathbf{X}_1)] = \mathbf{0}$ and $\mathbb{E}[\mathbf{s}(\mathbf{X}_1) \mathbf{s}(\mathbf{X}_1)^T] = \mathbf{G}$, whence $\mathbb{E}[\mathbf{c}(\mathbf{X}_1)] = \mathbf{0}$ and $\mathbb{E}[\|\mathbf{c}(\mathbf{X}_1)\|^2] = \text{tr}(\mathbf{G}^{-1}) < \infty$. The vectors $\mathbf{s}(\mathbf{X}_1), \mathbf{s}(\mathbf{X}_2), \dots$ being independent, identically distributed, centred and square integrable, the multivariate central limit theorem gives $n^{-\frac{1}{2}} \sum_{i=1}^n \mathbf{s}(\mathbf{X}_i) \rightsquigarrow N(\mathbf{0}, \mathbf{G})$, so that by (1.62), Slutsky's lemma and the continuous mapping theorem,

$$\sqrt{n} \hat{\mathbf{u}}_n \rightsquigarrow N(\mathbf{0}, \mathbf{G}^{-1} \mathbf{G} \mathbf{G}^{-1}) = N(\mathbf{0}, \mathbf{G}^{-1}),$$

which is (1.46): in the $g_{[A_0, s_0]}$ -orthonormal basis $e_1, \dots, e_d$ the covariance operator $G_{[A_0, s_0]}^{-1}$ is represented by the matrix G^{-1} .

Step 7: the Procrustes form. By Proposition 16, $e_n = \iota(\hat{u}_n)$ on $\mathcal{A}_n$, and $\mathbb{P}(\mathcal{A}_n) \rightarrow 1$. If ξ_n and η_n agree on events of probability tending to one and $\xi_n \rightsquigarrow \xi$, then $\eta_n \rightsquigarrow \xi$ as well, since for every bounded Lipschitz test function ψ one has $|\mathbb{E}\psi(\xi_n) - \mathbb{E}\psi(\eta_n)| \leq 2\|\psi\|_\infty \mathbb{P}(\mathcal{A}_n^c) \rightarrow 0$ by the portmanteau theorem. Applying this and the continuous mapping theorem to the linear isometry ι turns (1.46) into (1.47); the limit is carried by $\iota(T_{[A_0, s_0]}M) = H_{(A_0, s_0)}$, which by Proposition 1 has dimension d . $\square$

Corollary 21. The influence operator (1.44) satisfies the influence relation of [Sun et al.(2026)Sun, Lin, and Liu], Lemma III.3, namely

$$\mathbb{E} [\text{IF}(\mathbf{X}_1) \otimes s(\mathbf{X}_1; [A_0, s_0])] = \text{id}_{T_{[A_0, s_0]}M}. \quad (1.63)$$

Moreover it is the only influence operator with values in the span of the score for which (1.43) can hold.

Proof. In the coordinates of Step 6 of the previous proof, (1.63) reads

$$\mathbb{E} [c(\mathbf{X}_1)s(\mathbf{X}_1)^T] = G^{-1}\mathbb{E} [s(\mathbf{X}_1)s(\mathbf{X}_1)^T] = G^{-1}G = I_d.$$

For the uniqueness, an influence operator of the stated form is $\text{IF}'(x) = Cs(x)$ for some $d \times d$ matrix C , and (1.63), which [Sun et al.(2026)Sun, Lin, and Liu] show to be forced for regular asymptotically linear sequences, gives $CG = I_d$, that is $C = G^{-1}$. $\square$

Remark 22. Three comments on the content of Theorem 20.

First, (1.47) is the well posed replacement of the false naive statement discussed at the beginning of this section: it is not the raw error $\hat{A}_n - A_0$ that is asymptotically Gaussian, but the error measured after the optimal rotational alignment $\hat{A}_n \mapsto \hat{A}_n Q_n$, and the limit lives on the horizontal space $H_{(A_0, s_0)}$ only. Viewed inside $\mathbb{R}^{p \times q} \times \mathbb{R} \cong \mathbb{R}^{pq+1}$ the limiting Gaussian law is degenerate, carrying no mass in the $\frac{q(q-1)}{2}$ vertical directions of Proposition 1. This degeneracy is a feature and not a defect: those are exactly the directions along which the likelihood (1.36) is constant, so no data can ever be informative about them.

Second, (1.46) contains as a by-product the rate of convergence, since it gives $\left| \text{Exp}_{[A_0, s_0]}^{-1}([\hat{A}_n, \hat{s}_n]) \right| = O_{\mathbb{P}}(n^{-\frac{1}{2}})$ and hence, on $\mathcal{A}_n$, that the Riemannian distance between $[\hat{A}_n, \hat{s}_n]$ and $[A_0, s_0]$ is $O_{\mathbb{P}}(n^{-\frac{1}{2}})$. This is the quantitative refinement of Proposition 13.

Third, no assumption whatsoever has been made on the multiplicities of the singular values of A_0 . This is worth emphasizing, because the classical route to such a theorem, namely perturbation of the eigenvectors of S_n followed by the delta method, requires the leading q eigenvalues of Σ_0 to be simple. When they are not, individual principal directions are genuinely not asymptotically normal, and yet the point $[\hat{A}_n, \hat{s}_n]$ of M still is: the quotient level limit simply does not see the multiplicities. The reason is Theorem 7, where positive definiteness of $G_{[A_0, s_0]}$ was obtained from $V_{(A_0, s_0)} \cap H_{(A_0, s_0)} = \{0\}$ alone.

Remark 23. Theorem 20 is the input required by the optimality theory of [Sun et al.(2026)Sun, Lin, and Liu]. Together with the local asymptotic normality expansion of Proposition 8 it yields, through Le Cam's third lemma, that the maximum likelihood estimator is locally regular in the sense of their Definition III.5, and then their convolution theorem, Theorem III.1, and their local asymptotic minimax theorem, Theorem III.2, identify $G_{[A_0, s_0]}^{-1}$ as the efficiency bound over all regular estimator sequences; compare their Theorem V.1. Carrying this out requires the parallel transport of (M, g) and a comparison between the transported and the coordinate errors, and is left for the next section.

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